

# FALCON's Response to Amazon for Supply of Linear CBS 6K PPH

R1\_Option AMZ

October 3, 2024





Kind Attention –

**Mr. Rahul Didwani**

**Mr. Vinayak Garg**

**M/s Amazon**

**Offer Ref: F24-00403\_R1 Option AMZ; Date: 03-10-2024**

**Subject – Techno -Commercial Offer of Linear Cross Belt Sorter 6K PPH**

We are pleased to submit our Techno-Commercial Offer in response to your RFP.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

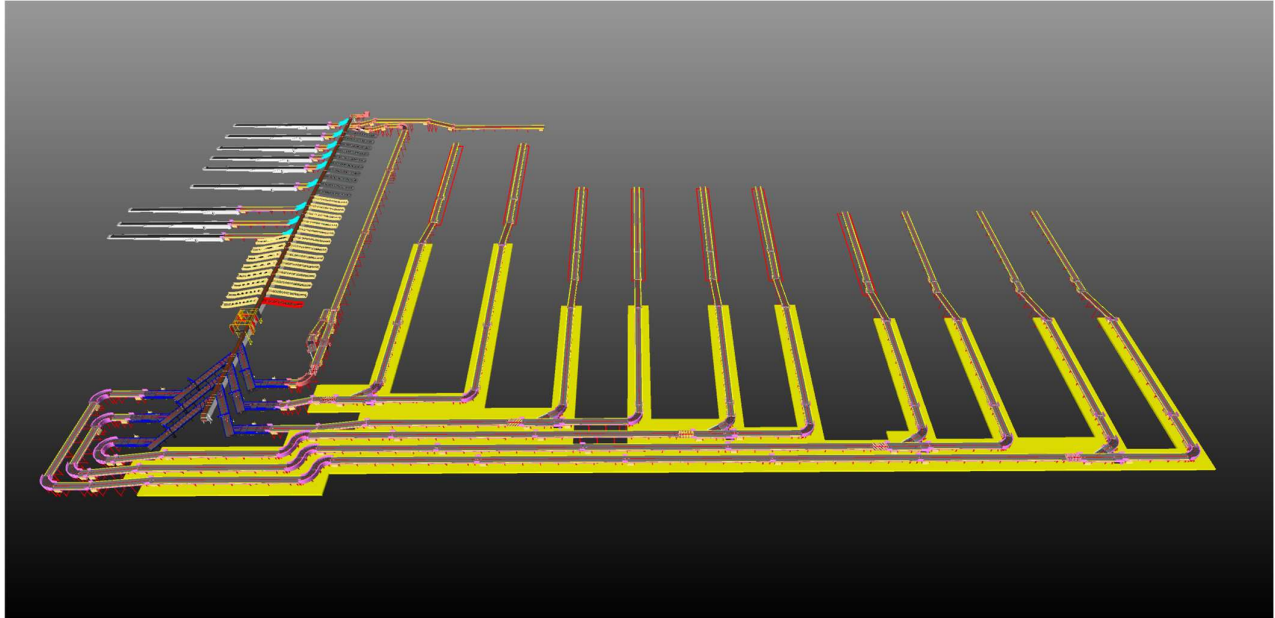
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

**Best Regards,**

**Sandeep Bansal**

**Chief Business Officer**

## Response to RFP for Linear Cross Belt Sorter 6K PPH



Proposal Reference – F24-00403-R1\_Op. AMZ

Falcon Autotech Private Limited  
Plot No. 87, Sector Ecotech-1, Extension-1,  
Greater Noida, Uttar Pradesh 201308.

Contact – Sanyog Pratap Singh  
Asst. Manager  
Mob - +91 8750052591  
Sanyog.singh@falconautoonline.com

## Contents

|                                                                                     |    |
|-------------------------------------------------------------------------------------|----|
| Contents.....                                                                       | 4  |
| 1. Glossary .....                                                                   | 7  |
| 2. Executive Summary .....                                                          | 8  |
| 3. Company Profile .....                                                            | 9  |
| 4. Falcon's Experience and Achievements in Sortation Space Globally .....           | 13 |
| 5. Reference Projects .....                                                         | 15 |
| 5.1 Project 1- (CEP Client, UK) .....                                               | 15 |
| 5.2 Project 2- (CEP Client, Riyadh) .....                                           | 16 |
| 5.3 Project 3- (Client – Courier Express Shipment, Dubai & Jeddah) .....            | 17 |
| 5.4 Project 4- (CEP Client, Sydney).....                                            | 18 |
| 5.5 Project 5- (Client – E-Commerce, India).....                                    | 19 |
| 5.6 Project 6- (Ecommerce Client, India) .....                                      | 20 |
| 6. Handled Shipment Spectrum .....                                                  | 21 |
| 6.1 Shipment size loadable on the sorter .....                                      | 21 |
| 6.2 Shipments to be loaded on Sorter shall have the following characteristics:..... | 21 |
| 6.3 Shipment not loadable on the sorter .....                                       | 22 |
| 6.4 Pictures of Sortable Shipments.....                                             | 23 |
| 7. Proposed System Description .....                                                | 24 |
| 7.1 Objective.....                                                                  | 24 |
| 7.2 Summary of the System .....                                                     | 24 |
| 7.3 Main benefits of the proposed solution.....                                     | 25 |
| 8. Concept Description .....                                                        | 26 |
| 9. Layout View .....                                                                | 27 |
| 10. Proposed System Capacity Calculations .....                                     | 28 |
| 10.1 Sorter System Capacity .....                                                   | 28 |
| 11. System description .....                                                        | 29 |
| 11.1 Infeed System .....                                                            | 29 |
| 11.1.1 Infeed System .....                                                          | 30 |
| 11.1.1.1 Straight and Inclined Powered Belt Conveyor .....                          | 31 |
| 11.1.1.2 ZTR Curves (Pic for ref.).....                                             | 33 |
| 11.1.1.3 Buffer Conveyor .....                                                      | 33 |
| 11.1.1.4 Angle Merge.....                                                           | 34 |
| 11.1.1.5 Roller Aligner conveyor.....                                               | 34 |
| 11.1.2 Conveyor BOQ .....                                                           | 35 |
| 11.2 Induction to Sorter .....                                                      | 38 |

|          |                                                                            |    |
|----------|----------------------------------------------------------------------------|----|
| 11.2.1   | <b>Feedlines</b> .....                                                     | 38 |
| 11.2.1.1 | Weighing Conveyor.....                                                     | 39 |
| 11.2.1.2 | Receiving Conveyor.....                                                    | 39 |
| 11.2.1.3 | Buffer Conveyors.....                                                      | 40 |
| 11.2.2   | <b>Linear Sorter</b> .....                                                 | 41 |
| 11.2.3   | <b>Sorting Output</b> .....                                                | 42 |
| 11.2.3.1 | Live Chute (Direct Loading Type) .....                                     | 42 |
| 11.2.3.2 | Collection Chute (Friction Roller Type) .....                              | 43 |
| 11.2.3.3 | Rejection Chute (Friction Roller Type) .....                               | 44 |
| 11.2.4   | <b>Recirculation &amp; Manual Feed Line</b> .....                          | 45 |
| 11.2.4.1 | Conveyor BOQ.....                                                          | 45 |
| 12.      | <b>Description of Components of Equipment</b> .....                        | 46 |
| 12.1     | <b>Cross Belt Sorter -</b> .....                                           | 46 |
| 12.1.1   | CBS Carrier .....                                                          | 46 |
| 12.1.2   | Servo Roller.....                                                          | 47 |
| 12.1.3   | Chassis.....                                                               | 47 |
| 12.1.4   | Carrier Wheels .....                                                       | 47 |
| 12.1.5   | Non-contact based linear motor drive .....                                 | 48 |
| 12.1.6   | Power Transmission .....                                                   | 48 |
| 12.1.7   | Data Transmission .....                                                    | 48 |
| 12.1.8   | Carriers positioning System. ....                                          | 49 |
| 12.1.9   | Technical specifications of Sorter .....                                   | 49 |
| 12.2     | <b>Scanning &amp; Sensing on Linear Sorter</b> .....                       | 50 |
| 12.2.1   | Automatic Barcode Scanning .....                                           | 50 |
| 12.2.2   | Shipment Positioning System. ....                                          | 50 |
| 12.3     | <b>Steel Works – Mezzanine &amp; Staircases- (In Amazon’s Scope)</b> ..... | 51 |
| 12.3.1   | Mezzanine.....                                                             | 51 |
| 12.3.2   | Staircase.....                                                             | 51 |
| 13.      | <b>Proposed System Technical Details</b> .....                             | 52 |
| 13.1     | <b>Mechanical equipment</b> .....                                          | 52 |
| 13.2     | <b>Electrical Equipment</b> .....                                          | 53 |
| 13.3     | <b>Control System</b> .....                                                | 53 |
| 14.      | <b>Electrical System</b> .....                                             | 53 |
| 14.1     | <b>Reference Picture of Power Distribution Panel</b> .....                 | 53 |
| 14.2     | <b>Main Control Panel (Reference)</b> .....                                | 54 |
| 14.3     | <b>Induct Stations Control Panel (Reference)</b> .....                     | 54 |

|      |                                                                           |    |
|------|---------------------------------------------------------------------------|----|
| 15.  | Controls Architecture (Ref.) .....                                        | 55 |
| 16.  | IT Architecture (Ref.) .....                                              | 56 |
| 16.1 | HAA Server (In Amazon Scope) .....                                        | 56 |
| 17.  | Falcon's WCS CONTROLIT .....                                              | 57 |
| 18.  | Falcon's Visual Inspection System (SCADA) .....                           | 58 |
| 19.  | Key Components Make .....                                                 | 58 |
| 20.  | Principle of Safety .....                                                 | 59 |
| 20.1 | E-Stops .....                                                             | 59 |
| 21.  | Program Organisation .....                                                | 60 |
| 21.1 | Program/ Project Schedule .....                                           | 60 |
| 21.2 | Program Management .....                                                  | 60 |
| 22.  | Amazon Responsibility .....                                               | 60 |
| 22.1 | Amazon Responsibilities During the Assembly and Commissioning Phase ..... | 60 |
| 22.2 | Responsibilities of Amazon During the Tests .....                         | 61 |
| 22.3 | Amazon Responsibilities During the Training .....                         | 61 |
| 23.  | Commercial Terms .....                                                    | 62 |
| 23.1 | Price Sheet .....                                                         | 62 |
| 23.2 | Payment Terms .....                                                       | 62 |
| 24.  | Warranty Period .....                                                     | 63 |
| 25.  | Exclusions .....                                                          | 64 |

## 1. Glossary

| S. No. | Term | Description                       |
|--------|------|-----------------------------------|
| 1      | RFP  | Request For Proposal              |
| 2      | PPH  | Shipments Per Hour                |
| 3      | ICR  | Intelligent Character Recognition |
| 4      | MEZZ | Mezzanine                         |
| 5      | LIM  | Linear Induction Motors           |
| 6      | AC   | Alternating Current               |
| 7      | DC   | Direct Current                    |
| 8      | PLC  | Programmable Logic Controller     |
| 9      | IT   | Information Technology            |
| 10     | BOQ  | Bill Of Quantity                  |
| 11     | I/O  | Input/ Output                     |
| 12     | PDP  | Power Distribution Panel          |
| 13     | PC   | Personal Computer                 |
| 14     | UPS  | Uninterrupted Power Supply        |
| 15     | CBS  | Cross Belt Sorter                 |
| 16     | MDR  | Motor Driven Roller               |
| 17     | DAP  | Design Approval Phase             |

## 2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this Amazon RFQ of Linear CBS. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success. Following the same objective for the CBS system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following key characteristics:

- **Linear CBS for Parcel sorting, employing cross belt technology, have been designed to handle a throughput of 6,000 packages per hour.**
- **This Linear CBS is equipped with 6 Nos Fully Automatic induct lines, Along with manual loading point.**
- **The System is having its own conveyor connection to transport the volume from primary loading points to the induction zone.**
- **In the system, there are 9 Live chutes (connect to TBC), 20 Nos Collection Chute, 1 rejection chute, 1 RC Line, 1 recirculation line & 1 Sortfail line..**
- **The system layout has been meticulously planned to facilitate smooth operational flow, ensuring efficient movement of personnel.**

**A tailor-made and simple layout designed explicitly for Amazon.** The proposed layout is the result of the technical requirements in the RFP document and our discussions with the relevant stakeholders during our meeting.

### 1. **Falcon's reliable Shipment sortation systems**

These systems are globally being used by most innovative brands such as Amazon, Flipkart, Delhivery, Asendia, Fastway and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

### 2. **Commitment to quality systems**

Demonstrating Falcon's clear commitment to the Amazon satisfaction, the shipment sortation system, parts and services will be under warranty for 12 months from installation go-live.

### 3. **A strong and already successful partnership between Amazon & Falcon**

Amazon has been Falcon's customer since long and using multiple Falcon Systems. For each project, several advanced tools to improve the service's quality and efficiency are provided: Service Desk, Hotline support, remote access, regular governance meetings and monthly reports.



### 3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

## Falcon 2.0

Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

**5**  
Product Lines

**15+**  
Countries with Live Installations

**1800+**  
Total Installed Systems Globally

**1Million+**  
Sorts Per Hour

**600+**  
Employees

#### Long-standing Relationships with industry leaders

#### News & Articles

**FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING**  
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

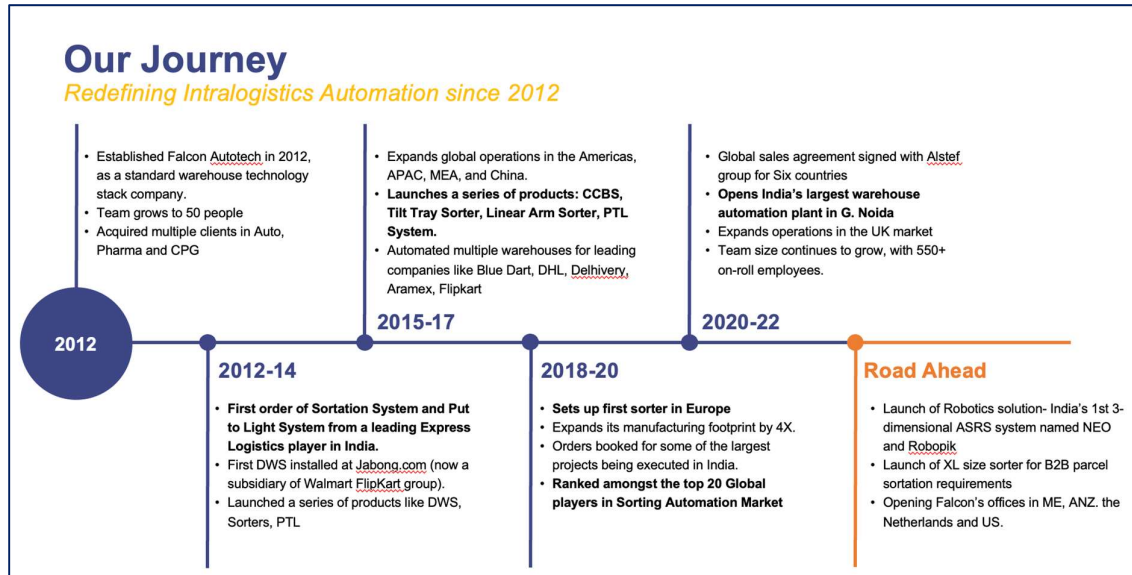
**FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP**  
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions  
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.

### Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



## Product and Solutions

With 100% focused on Parcels, Eaches, Totes, Bags, Cartons



### SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- Popup Sorter
- Sweep Sorter
- Pusher Sorter



### PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



### DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
- R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
- Mini (600MM)
- Jumbo (1200MM)



### CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



### ROBOTICS

- NEO
- Robopick

Powered by

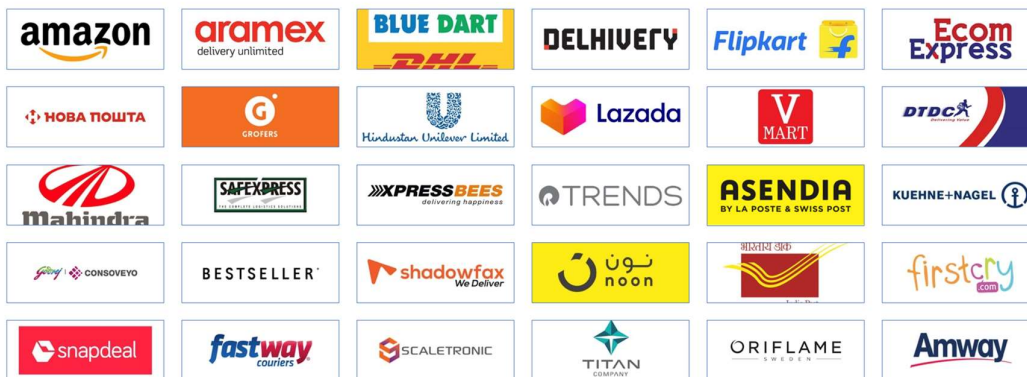


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

## Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements



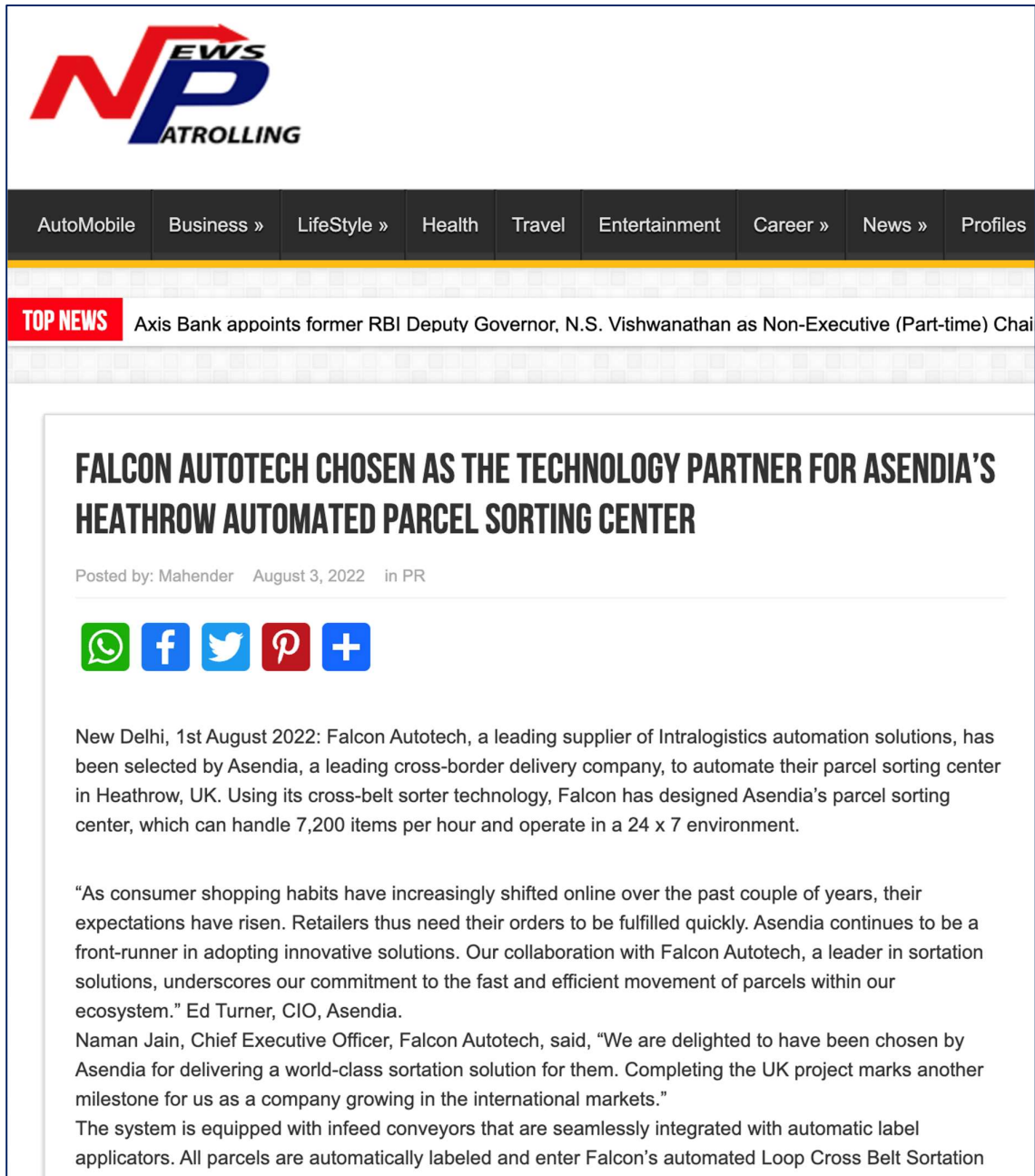
With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.





#### 4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of the **World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



POSTED ON APRIL 17, 2023 IN / NEWS

## Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17<sup>th</sup> April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

## Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

## 5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

### 5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Linear Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

#### **Solution Specifications –**

- Throughput: 7200 PPH
- End Destinations: 58 Nos

#### **Key Technology Modules –**

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- WCS Software System.

#### **Site Picture –**



## 5.2 Project 2- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of shipments arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

### Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

### Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Shipment handling.
- FOCR Engine.
- WCS Software System.

### Site Picture –





### 5.3 Project 3- (Client – Courier Express Shipment, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 shipments per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

#### **Solution Specifications –**

- Throughput: 3000 PPH
- End Destinations: 480 Nos

#### **Key Technology Modules –**

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

#### **Site Picture –**



#### 5.4 Project 4- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Linear Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

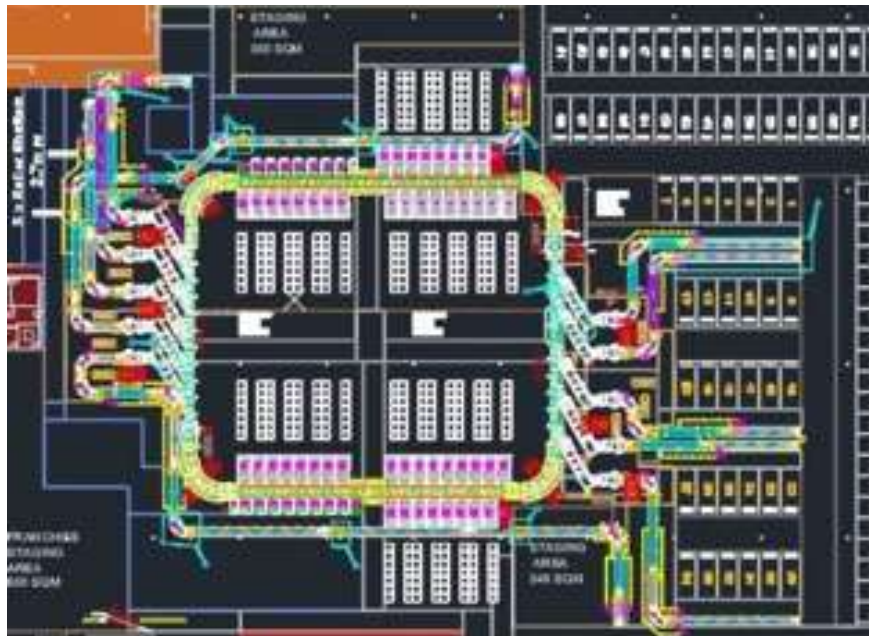
##### **Solution Specifications –**

- Throughput: 16000 PPH
- End Destinations: 369 Nos

##### **Key Technology Modules –**

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Linear Cross Belt Sorter.
- WCS Software System.

##### **Site Picture –**



### 5.5 Project 5- (Client – E-Commerce, India)

#### Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

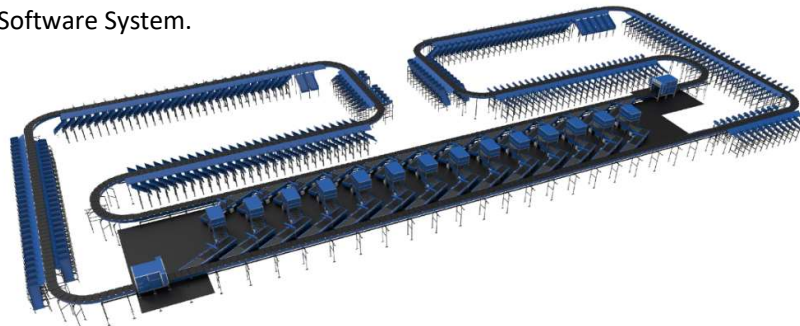
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

#### Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

#### Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



## 5.6 Project 6- (Ecommerce Client, India)

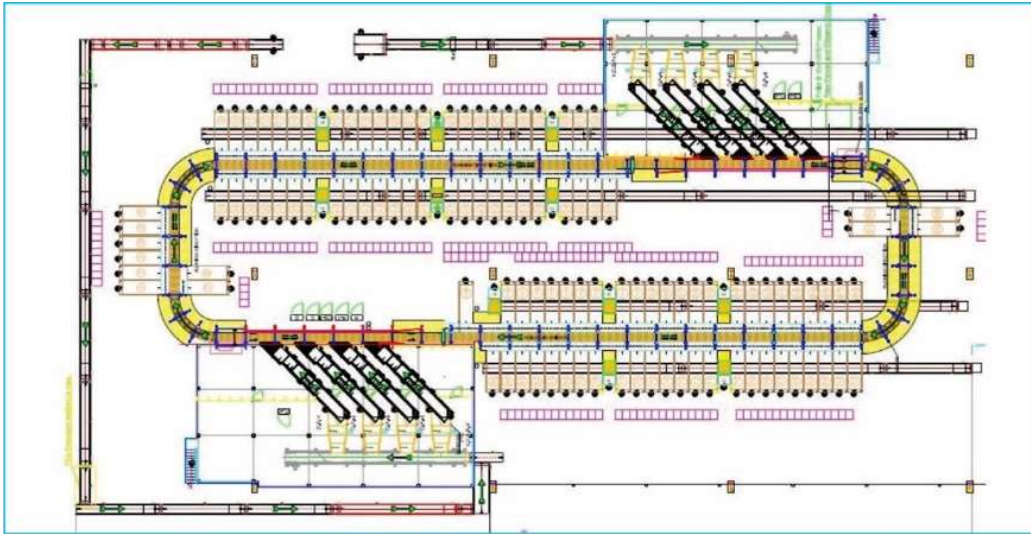
Use Case - Destination Sorting of Packed Shipments

### Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

**Key Technology Modules** - Linear Cross Belt Sorter with a cell size of 900 x 500 mm

### Layout and Site Pictures -





## 6. Handled Shipment Spectrum

As per the shipment spectrum data provided in the RFP documents, Falcon has studied and analysed the shipment spectrum in detailed.

Falcon purposes to use its “Linear Cross Belt Sorter” to provide the maximum benefits to Amazon in-terms of handling variety of shipment sizes and weight.

### 6.1 Shipment size loadable on the sorter

Falcon’s Cross belt sorter has a capability to handle the Totes & Boxes of below mentioned sizes and weight.

| Specification | Unit | Value |
|---------------|------|-------|
| Max Length    | mm   | 800   |
| Max Width     | mm   | 800   |
| Max Height    | mm   | 700   |
| Max Weight    | Kg   | 30    |
| Min length    | mm   | 100   |
| Min Width     | mm   | 100   |
| Min Height    | mm   | 50    |
| Min Weight    | gm   | 200   |

### 6.2 Shipments to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of shipment cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Shipments shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The shipments must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

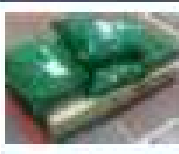
### 6.3 Shipment not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile shipments with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.



## 6.4 Pictures of Sortable Shipments

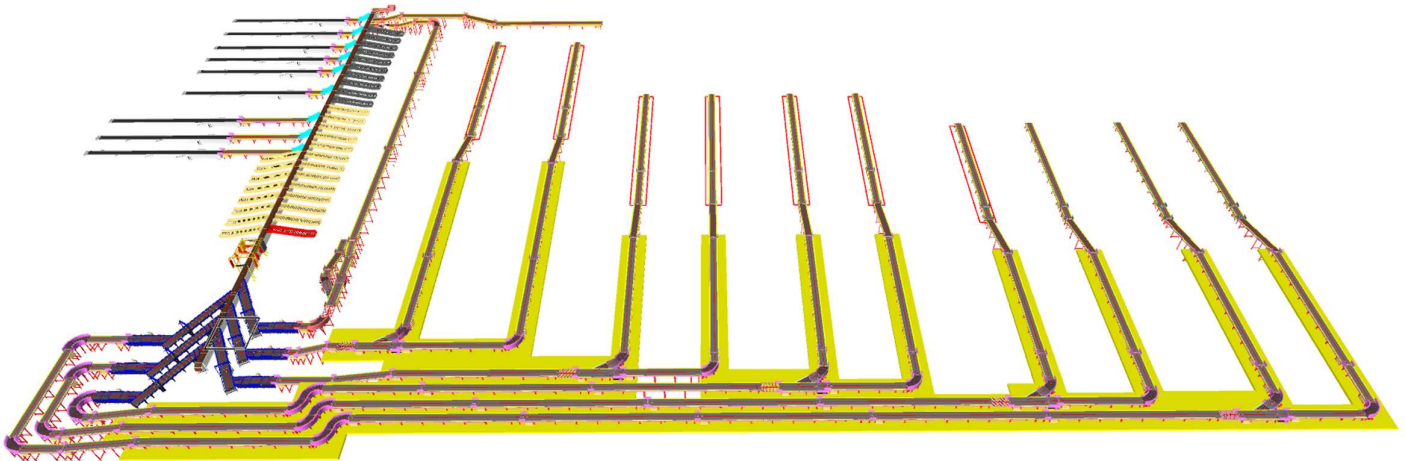
| PARCEL TYPE                                                                         |                                                    | DESCRIPTION                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Strongly Strapped                                  | A parcel banded with strapping material.                                                                                                                                                                              |
|    | Pre Manufactured Plastic Courier Pack <A3 in size  | A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.                                                                                                                        |
|    | Pre Manufactured Plastic Courier Pack > A3 in size | A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (30 m) may create problems and are not recommended to be processed. |
|    | Other Plastic Covered Item                         | A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.                                                                                                                 |
|    | Paper Covered Item                                 | A cardboard parcel coated in a packing paper.                                                                                                                                                                         |
|   | Cloth Covered Item                                 | Often import parcels that are of irregular shape.                                                                                                                                                                     |
|  | Wine Box                                           | A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.                                                                                        |
|  | Cardboard Box                                      | A cardboard parcel is often the norm.                                                                                                                                                                                 |
|  | Totes                                              | A large plastic, lidded box.                                                                                                                                                                                          |
|  | Flats                                              | A parcel that often contains documents.                                                                                                                                                                               |
|  | Poly Bags                                          | It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.                                                                                                      |

## 7. Proposed System Description

### 7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Conveyor Automation & Linear CBS for sorting Totes & Boxes as per Amazon's logic.

### 7.2 Summary of the System



Process flow of the Linear CBS System-

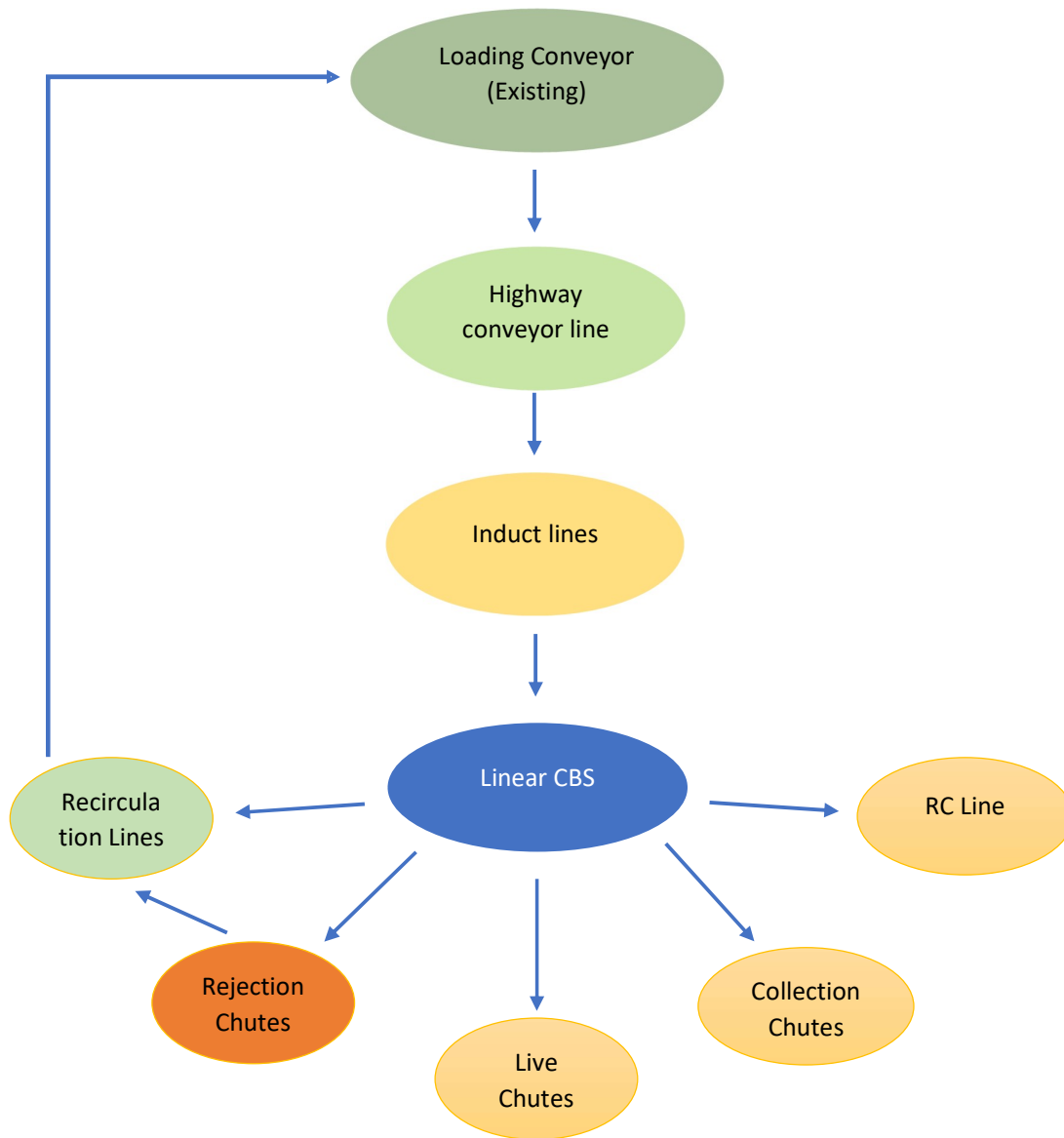
1. **Infeed System:** - Boxes and totes are loaded onto the existing conveyor in a lengthwise orientation. From there, they are directed to their assigned highway line, which transports them to the CBS induct zone in a singulated manner.
2. **Inducts:** - Upon arrival at the induct zone, Falcon's fully automatic induct line accurately and smoothly inducts the parcels onto the Linear CBS, based on their dimensions and weight.
3. **Linear CBS:** - Once the parcels enter the main Linear CBS, the Cross-Belt Sorter (CBS) capture the barcode details & volume data after which it efficiently sorts the boxes and totes into their designated output chutes using data provided by Amazon.
4. **Output Chutes:** - The Totes/Boxes are discharged into below output chutes.
  - a. **Live Chutes** - There are 9 sliding-type live chutes within the Linear CBS system, integrated with PVC belt conveyors and TBCs for live loading.
  - b. **Collection chute** – A total of 20 friction roller-based chutes are designed to collect and gradually accumulate the parcels.
  - c. **Rejection Chute**- One friction roller-based chute handles rejected shipments.
5. **Recirculation Line:** - A recirculation line is available to automatically feed sortfail parcels back into the Linear CBS. It is also integrated with a manual loading point for reprocessed boxes and totes collected from the rejection chute.

### 7.3 Main benefits of the proposed solution

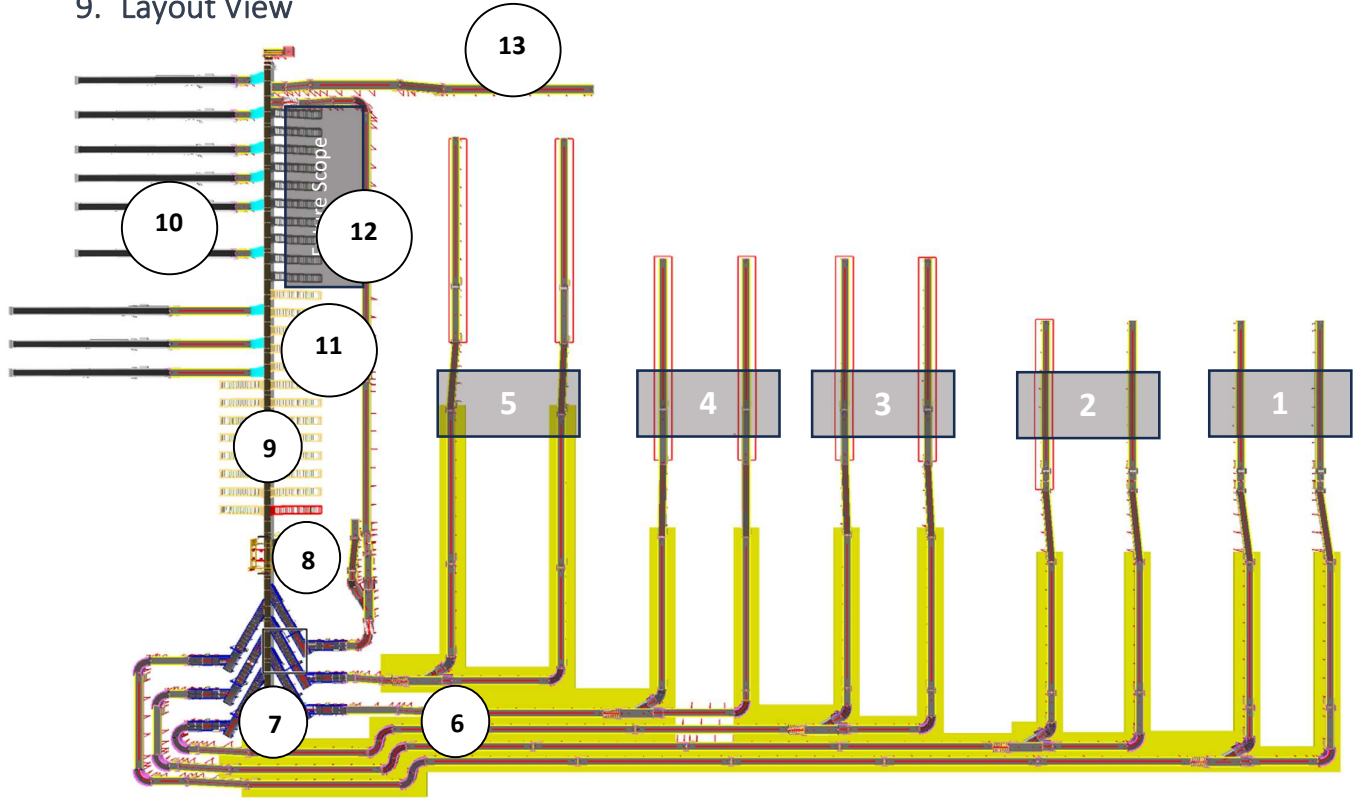
The proposed solution has the following advantages:

1. High operational throughput
2. Good Tote/Box conveying quality for the entire range of conveyable Shipments with a sorting conveyor speed of up to 1 m/s.
3. Low occupancy of floor space in the building.
4. Narrow discharge centers for the increased number of splits in limited space
5. FALCON's CBS can adapt to changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

## 8. Concept Description



## 9. Layout View

**Legend**

- 1 – Loading Conveyor – Zone 1
- 2 – Loading Conveyor – Zone 2
- 3 – Loading Conveyor – Zone 3
- 4 – Loading Conveyor – Zone 4
- 5 – Loading Conveyor – Zone 5
- 6 – Highway Lines
- 7 – Induct Zone
- 8 – ICR Tunnel
- 9 – Main Linear CBS
- 10 – Live Chutes
- 11 – Collection Chute
- 12 – Recirculation Line
- 13 – RC Line

## 10. Proposed System Capacity Calculations

### 10.1 Sorter System Capacity

The following table shows the throughput calculation for the sortation system designed based on Amazon requirements.

| SPECIFICATION                                | VALUE                                      |
|----------------------------------------------|--------------------------------------------|
| Sorter Type                                  | Linear CBS                                 |
| Speed                                        | 1.5 m/s                                    |
| Pitch                                        | 500 mm                                     |
| Carrier per hour                             | 10,800 CPH                                 |
| Belts per hour                               | 10,800 BPH                                 |
| Designed Throughput of 1 Linear CBS (A)      | 10,800 Boxes per hour                      |
| Per Feedline Capacity (Single Worthy Parcel) | 3000 Boxes per hour                        |
| No of Feedlines                              | 5 Nos.                                     |
| Total Feedline designed capacity (B)         | 15000 Boxes per hour                       |
| System designed throughput (Lower of A&B)    | 10,800 Boxes per hour (Single Belt Worthy) |

CBS Throughput Based on the percentage of Dual Belt worthy Parcels.

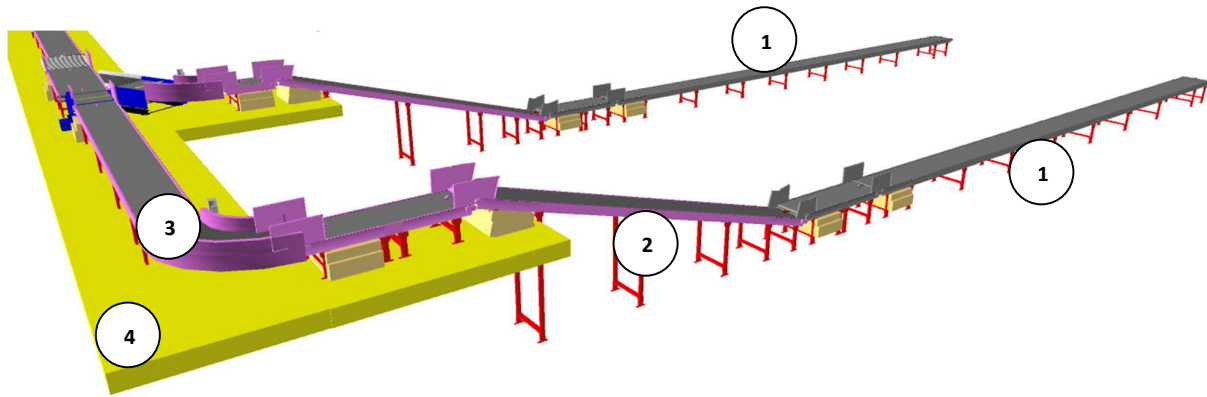
| SPECIFICATION                   | VALUE      |
|---------------------------------|------------|
| 100% of Dual belt worthy Parcel | 5,400 PPH  |
| 50% of Dual belt worthy Parcel  | 7,560 PPH  |
| 00% of Dual belt worthy Parcel  | 10,800 PPH |

- Any Parcel dimension greater than 400X600 will be considered as Dual belt worthy Parcel.



## 11. System description

### 11.1 Infeed System



The system is equipped with three infeed conveyors, facilitating the movement of shipments from the lower level to the mezzanine level.

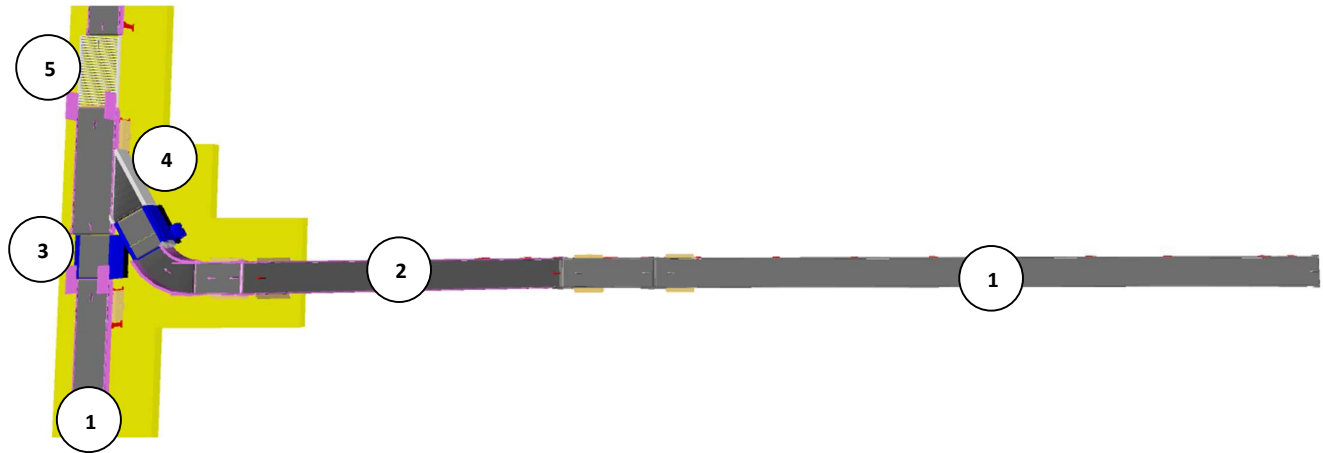
#### Legend

- 1- Manual loading lines (Existing)
- 2- Inclined Conveyor
- 3- Highway Conveyor
- 4- Mezzanine

### 11.1.1 Infeed System

Infeed conveyor system consists of below components:

1. Straight Belt Conveyor
2. Inclined PVC Conveyor
3. Buffer Conveyor
4. Angle Merge
5. Side Aligner



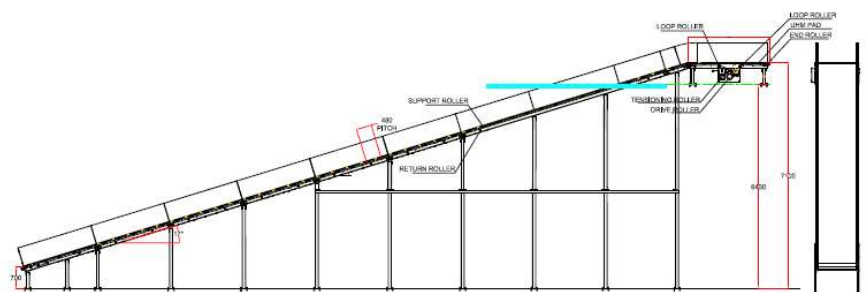
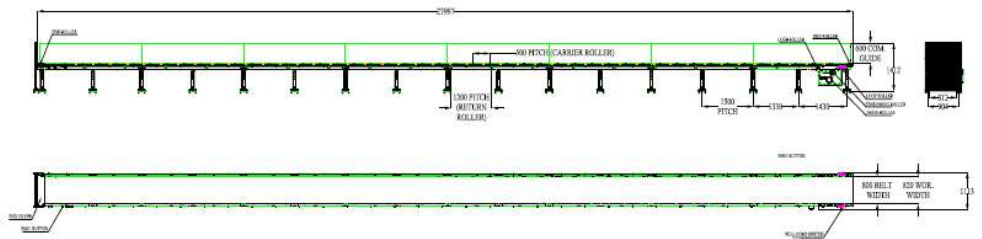
#### 11.1.1.1 Straight and Inclined Powered Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

### Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



| Specification                             | UOM  | Remark                                                                                                               |
|-------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------|
| Manufacturer Name                         | Name | Falcon Autotech                                                                                                      |
| Material of Roller                        | Type | MS Rollers with Zinc Plating                                                                                         |
| MOC of Belt                               | Type | PVC                                                                                                                  |
| Belt Thickness                            | mm   | 3                                                                                                                    |
| Belt Finish                               | Type | Matt Finish                                                                                                          |
| Belt width                                | mm   | 1000                                                                                                                 |
| Belt Joint Type                           | Type | Vulcanized, Endless Belts                                                                                            |
| Belt Make/Model No.                       | Make | FORBO/DERCO                                                                                                          |
| Chassis                                   | mm   | MS, 3mm Thick, Powder Coated-75 Microns                                                                              |
| Idler Roller Pitch on running side (Top)  | mm   | 480                                                                                                                  |
| Idler Roller Specs                        | Spec | Dia.50.8, shaft-12 mm, bearing-6301                                                                                  |
| End Rollers                               | Spec | Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns                    |
| Drive Rollers                             | Spec | OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING |
| Motor Shaft                               | MOC  | EN9 Shaft material                                                                                                   |
| Conveyor Under guarding                   | Spec | MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight                |
| Load capacity per unit length of Conveyor | kg/m | 50                                                                                                                   |
| Type of Guides                            | Type | Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.           |
| Drive Power Rating                        | kW   | Motor Rating depends on Length of the Conveyor                                                                       |
| Type of Motor                             | Type | AC Geared Motor                                                                                                      |
| Ingress Protection                        | Type | IP 55                                                                                                                |
| Type of Drive System                      | Type | Direct Shaft Mounted Motor/ Flange Mounted                                                                           |
| Type of Drive                             | Type | Chain drive/Torque Arm type                                                                                          |
| Gear Motor                                | make | Sew/Nord                                                                                                             |
| Drive                                     | make | Lenze/Siemens/Omron/Allen Bradley                                                                                    |
| Insulation Class                          | Type | Yes                                                                                                                  |
| Conveyor Speed                            | m/s  | Variable speed to meet TPH requirements                                                                              |
| Type of Mounts                            | Type | Fixed Heights Legs with Grouting Provisions                                                                          |

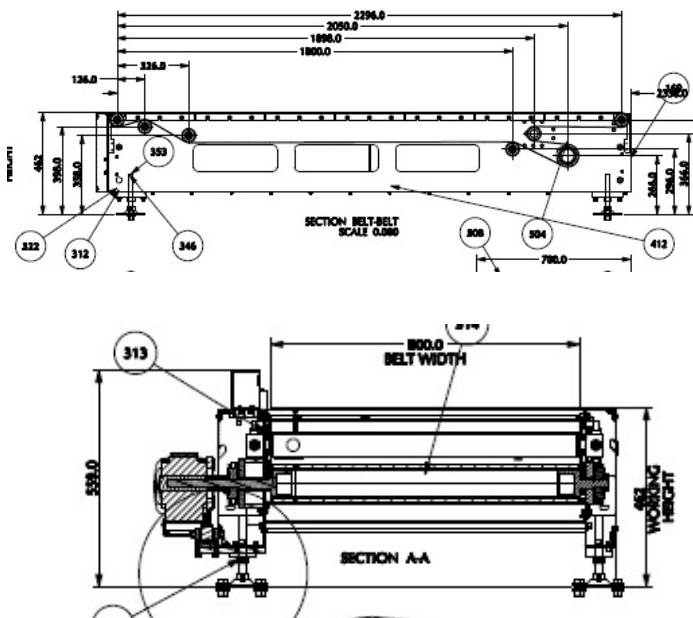
#### 11.1.1.2 ZTR Curves (Pic for ref.)



#### 11.1.1.3 Buffer Conveyor

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.



#### 11.1.1.4 Angle Merge

In the proposed solution this is 30° triangular high-speed conveyor used for inducting shipment/boxes directly on to the sorter. The Belts are Strip Belts for smooth shipment movement.



#### 11.1.1.5 Roller Aligner conveyor





11.1.2 Conveyor BOQ

| S No. | Name              | EL_1 | EL_2 | Length (m) | width (mm) | MP (KW) | Set |
|-------|-------------------|------|------|------------|------------|---------|-----|
| 1     | PVC Belt Conveyor | 698  | 3080 | 9          | 1010       | 3       | 1   |
| 2     | PVC Belt Conveyor | 3000 | 3000 | 19         | 1010       | 4       | 1   |
| 3     | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 4     | PVC Belt Conveyor | 3000 | 3000 | 5          | 1010       | 1.5     | 1   |
| 5     | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 6     | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 7     | PVC Belt Conveyor | 1700 | 1700 | 12         | 1010       | 3       | 1   |
| 8     | PVC Belt Conveyor | 3001 | 3001 | 20         | 1010       | 4       | 1   |
| 9     | PVC Belt Conveyor | 2904 | 1813 | 7          | 1010       | 3       | 1   |
| 10    | PVC Belt Conveyor | 698  | 3080 | 9          | 1010       | 3       | 1   |
| 11    | PVC Belt Conveyor | 3000 | 3000 | 17         | 1010       | 3       | 1   |
| 12    | PVC Belt Conveyor | 3000 | 3000 | 5          | 1010       | 1.5     | 1   |
| 13    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 14    | PVC Belt Conveyor | 3000 | 3000 | 16         | 1010       | 3       | 1   |
| 15    | PVC Belt Conveyor | 698  | 3080 | 9          | 1010       | 3       | 1   |
| 16    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 17    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 18    | PVC Belt Conveyor | 698  | 3081 | 9          | 1010       | 3       | 1   |
| 19    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 20    | PVC Belt Conveyor | 3000 | 3000 | 9          | 1010       | 2.2     | 1   |
| 21    | PVC Belt Conveyor | 3000 | 3000 | 4          | 1210       | 1.5     | 1   |
| 22    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 23    | PVC Belt Conveyor | 698  | 3080 | 9          | 1010       | 3       | 1   |
| 24    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 25    | PVC Belt Conveyor | 3000 | 3000 | 12         | 1010       | 3       | 1   |
| 26    | PVC Belt Conveyor | 3000 | 3000 | 4          | 1210       | 1.5     | 1   |
| 27    | PVC Belt Conveyor | 3000 | 3000 | 12         | 1010       | 3       | 1   |
| 28    | PVC Belt Conveyor | 3000 | 3000 | 13         | 1010       | 3       | 1   |
| 29    | PVC Belt Conveyor | 2898 | 1809 | 4          | 1010       | 2.2     | 1   |
| 30    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 31    | PVC Belt Conveyor | 698  | 3080 | 9          | 1010       | 3       | 1   |
| 32    | PVC Belt Conveyor | 2904 | 1813 | 7          | 1010       | 3       | 1   |
| 33    | PVC Belt Conveyor | 1700 | 1700 | 7          | 1010       | 1.5     | 1   |
| 34    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 35    | PVC Belt Conveyor | 698  | 3080 | 9          | 1010       | 3       | 1   |
| 36    | PVC Belt Conveyor | 3000 | 3000 | 3          | 1010       | 1.5     | 1   |
| 37    | PVC Belt Conveyor | 3000 | 3000 | 12         | 1010       | 2.2     | 1   |
| 38    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 39    | PVC Belt Conveyor | 3000 | 3000 | 4          | 1210       | 1.5     | 1   |
| 40    | PVC Belt Conveyor | 3000 | 3000 | 9          | 1010       | 2.2     | 1   |
| 41    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |
| 42    | PVC Belt Conveyor | 698  | 3080 | 9          | 1010       | 3       | 1   |
| 43    | PVC Belt Conveyor | 3000 | 3000 | 20         | 1010       | 4       | 1   |

|    |                        |      |      |    |      |     |   |
|----|------------------------|------|------|----|------|-----|---|
| 44 | PVC Belt Conveyor      | 3000 | 3000 | 20 | 1010 | 4   | 1 |
| 45 | PVC Belt Conveyor      | 3000 | 3000 | 4  | 1010 | 1.5 | 1 |
| 46 | PVC Belt Conveyor      | 3000 | 3000 | 4  | 1210 | 1.5 | 1 |
| 47 | PVC Belt Conveyor      | 3000 | 3000 | 10 | 1010 | 2.2 | 1 |
| 48 | PVC Belt Conveyor      | 3000 | 3000 | 17 | 1010 | 3   | 1 |
| 49 | PVC Belt Conveyor      | 698  | 3080 | 9  | 1010 | 3   | 1 |
| 50 | PVC Belt Conveyor      | 3000 | 3000 | 15 | 1010 | 3   | 1 |
| 51 | PVC Belt Conveyor      | 2905 | 1814 | 7  | 1010 | 3   | 1 |
| 52 | PVC Belt Conveyor      | 2906 | 1814 | 8  | 1010 | 3   | 1 |
| 53 | PVC Belt Conveyor      | 3000 | 3000 | 20 | 1010 | 4   | 1 |
| 54 | PVC Belt Conveyor      | 3000 | 3000 | 20 | 1010 | 4   | 1 |
| 55 | PVC Belt Conveyor      | 3000 | 3000 | 20 | 1010 | 4   | 1 |
| 56 | PVC Belt Conveyor      | 3000 | 3000 | 4  | 1010 | 1.5 | 1 |
| 57 | PVC Belt Conveyor      | 1700 | 1700 | 5  | 1010 | 1.5 | 1 |
| 58 | PVC Belt Conveyor      | 1700 | 1700 | 5  | 1010 | 1.5 | 1 |
| 59 | PVC Belt Conveyor      | 1701 | 1701 | 6  | 1010 | 1.5 | 1 |
| 60 | PVC Belt Conveyor      | 3000 | 3000 | 4  | 1210 | 1.5 | 1 |
| 61 | PVC Belt Conveyor      | 3000 | 3000 | 9  | 1010 | 2.2 | 1 |
| 62 | PVC Belt Conveyor      | 3000 | 3000 | 19 | 1010 | 4   | 1 |
| 63 | PVC Belt Conveyor      | 698  | 3080 | 9  | 1010 | 3   | 1 |
| 64 | PVC Belt Conveyor (S3) | 1700 | 1700 | 1  | 1010 | NA  | 1 |
| 65 | PVC Belt Conveyor (S3) | 3000 | 3000 | 2  | 1010 | NA  | 1 |
| 66 | PVC Belt Conveyor (S3) | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 67 | PVC Belt Conveyor (S3) | 1700 | 1700 | 1  | 1010 | NA  | 1 |
| 68 | PVC Belt Conveyor (S3) | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 69 | PVC Belt Conveyor (S3) | 1701 | 1701 | 1  | 1010 | NA  | 1 |
| 70 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 71 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 72 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 73 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 74 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 75 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 76 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 77 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 78 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 79 | Buffer Conveyor        | 3000 | 3000 | 1  | 1010 | NA  | 1 |
| 80 | Belt Merge 30°         | 3000 | 3000 | 0  | 1010 | NA  | 1 |
| 81 | Belt Merge 30°         | 3000 | 3000 | 0  | 1010 | NA  | 1 |
| 82 | Belt Merge 30°         | 3000 | 3000 | 0  | 1010 | NA  | 1 |
| 83 | Belt Merge 30°         | 3000 | 3000 | 0  | 1010 | NA  | 1 |
| 84 | Belt Merge 30°         | 3000 | 3000 | 0  | 1010 | NA  | 1 |
| 85 | Belt Turn 90°          | 1700 | 1700 | 0  | 1010 | NA  | 1 |
| 86 | Belt Turn 90°          | 1700 | 1700 | 0  | 1010 | NA  | 1 |
| 87 | Belt Turn 90°          | 1700 | 1700 | 0  | 1010 | NA  | 1 |
| 88 | Belt Turn 90°          | 1701 | 1701 | 0  | 1010 | NA  | 1 |
| 89 | Belt Turn 90°          | 3000 | 3000 | 0  | 1010 | NA  | 1 |

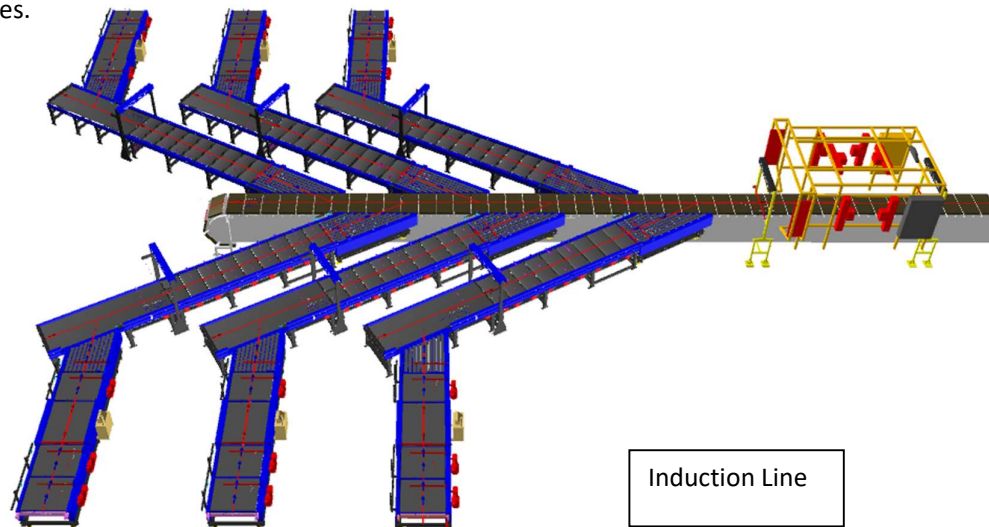
|     |                   |      |      |   |      |    |   |
|-----|-------------------|------|------|---|------|----|---|
| 90  | Belt Turn 90°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 91  | Belt Turn 90°     | 3001 | 3001 | 0 | 1010 | NA | 1 |
| 92  | Belt Turn 90°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 93  | Belt Turn 90°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 94  | Belt Turn 90°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 95  | Belt Turn 90°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 96  | Belt Turn 90°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 97  | Belt Turn 90°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 98  | Belt Turn 90°     | 3001 | 3001 | 0 | 1010 | NA | 1 |
| 99  | Belt Turn 90°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 100 | Belt Turn 90°     | 1700 | 1700 | 0 | 1010 | NA | 1 |
| 101 | Belt Turn 90°     | 1700 | 1700 | 0 | 1010 | NA | 1 |
| 102 | Belt Turn 60°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 103 | Belt Turn 60°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 104 | Belt Turn 60°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 105 | Belt Turn 60°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 106 | Belt Turn 60°     | 3000 | 3000 | 0 | 1010 | NA | 1 |
| 107 | Aligning Conveyor | 3000 | 3000 | 2 | 1200 | NA | 1 |
| 108 | Aligning Conveyor | 3000 | 3000 | 2 | 1200 | NA | 1 |
| 109 | Aligning Conveyor | 3000 | 3000 | 2 | 1200 | NA | 1 |
| 110 | Aligning Conveyor | 3000 | 3000 | 2 | 1200 | NA | 1 |
| 111 | Aligning Conveyor | 3000 | 3000 | 2 | 1200 | NA | 1 |

## 11.2 Induction to Sorter

The main purpose of the feedlines is to receive the Tote/Boxes from the infeed highway lines & induct it Smoothly & accurately onto the sorter by means of receiving conveyors, Weighing, Buffer/Spacing Conveyor and Angle Merge conveyors. The flow of shipments on the Inducts are controlled by sensors in such a way that the required gap is ensured between the Tote/Boxes.

Weighing Scales installed on the feedlines will capture the weight of each shipment and will perform weight control to detect out-of-spectrum shipments.

The system's design ensures that the maximum capacity of each feed, considered as the sum of the flows from the connected inductions, is always less than the planned induction capacity of 2500 for Totes & Boxes.



### 11.2.1 Feedlines

**Linear CBS has three fully Automatic Feedlines with Designed IPP of 3000 PPH.**

Each Feedline has 16 Conveyor Modules-

- Weighing Conveyor- 1 Nos.
  - Receiving Conveyor- 1 Nos.
  - Spacing Conveyor- 2 Nos.
  - Short Buffer conveyor- 10 Nos.
  - Angle Merge- 2 Nos.
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.
  3. Over the feedline, shipments are evenly spaced and smoothly transferred to the main Linear CBS.

#### 11.2.1.1 Weighing Conveyor

A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

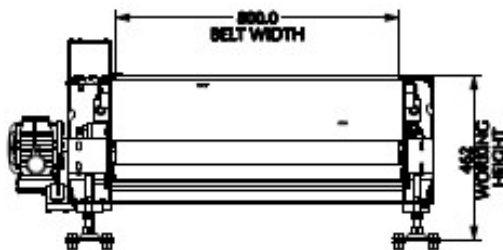
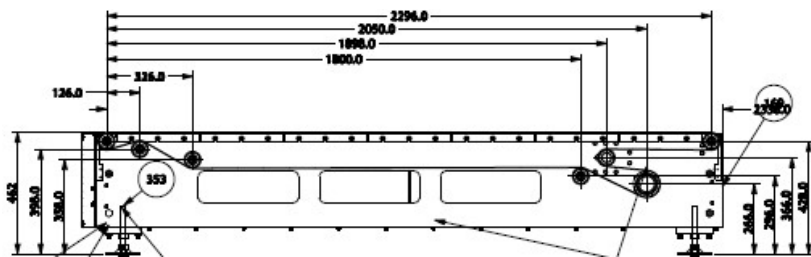
Weighing Conveyors equipped with high precision Load Cells to capture the weight of Shipments.

**Make- Bizerba/ Mettler Toledo**



#### 11.2.1.2 Receiving Conveyor

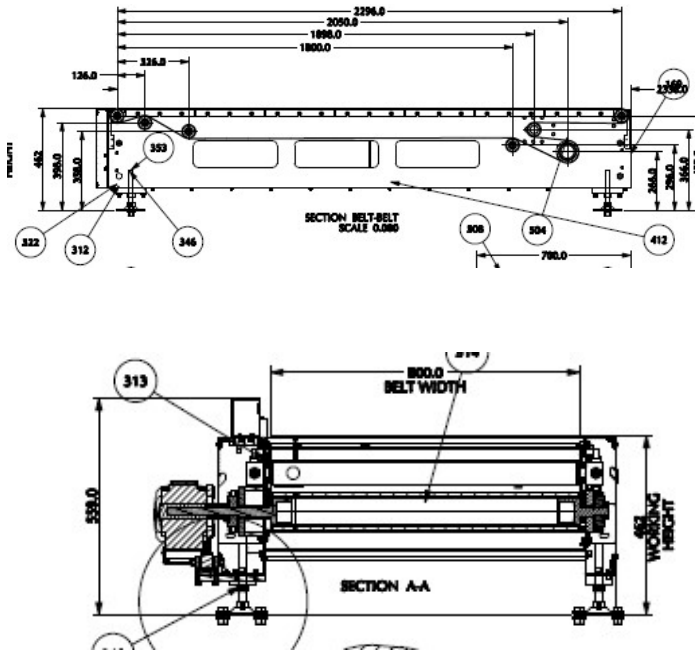
A Receiving conveyor is a type of conveyor system used to receive & Release the products for induction onto CBS. It serves as the connection point at turn point of entry where products are collected and conveyed to subsequent stages of the process.



### 11.2.1.3 Buffer Conveyors

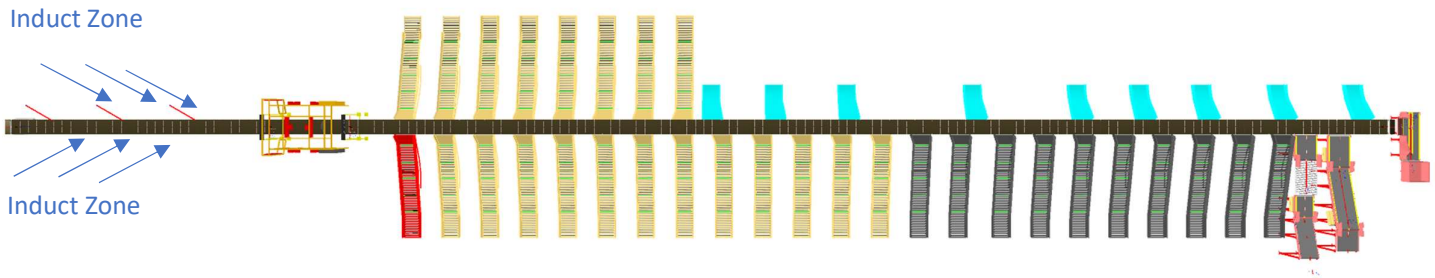
A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.





### 11.2.2 Linear Sorter



1. The proposed Linear CBS Sorter will be installed on ground level. The Sorter runs at 1700 mm from ground level.
2. We have provided Linear with single belt Carriers with 500 mm pitch and Belt Size of 495 x 800mm.
3. One Linear CBS has a top running length of 81 m & a total length of 164.5 m & 333 number of carriers.
4. As Parcel pass through the Barcode Scanning Tunnel on the Linear CBS, their barcodes are scanned, and chutes are assigned. Once the Parcel reaches the respective chute, the carrier carrying the Parcel for the chute will actuate and drop the Parcel in the assigned chute.

### 11.2.3 Sorting Output

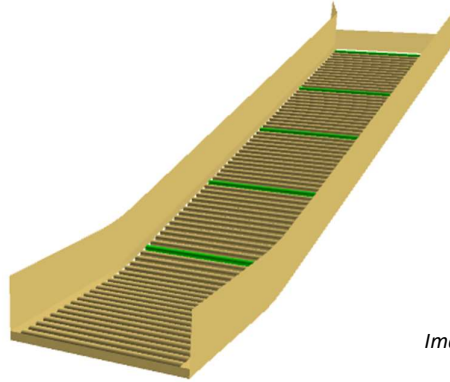
#### 11.2.3.1 Live Chute (Direct Loading Type)

A live Chute refers to a combination of collection chute, PVC belt conveyor & TBC (In Amazon's Scope) , where the Collection chute helps bringing down the sorted Parcel & release it to running conveyor for direct loading into the trucks.



### 11.2.3.2 Collection Chute (Friction Roller Type)

A Friction roller chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes its roller platform to gradually descend & collect the parcel at the end of the.



*Image for representation purpose only*

**Calculation for Design:** We have considered an average shipment size of 400 X 400 X 400 mm while designing the capacity of chutes.

| Average shipment size considered for chute volume calculation | L(m) | B(m) | H(m) | Volume (m3) |
|---------------------------------------------------------------|------|------|------|-------------|
|                                                               | 0.4  | 0.4  | 0.4  | 0.064       |
| Collection type chute                                         | 5.6  | 1    | 0.3  | 1.68        |

Volume Wise - Assuming a 100% occupancy rate, about **26 Parcel** will readily fit into these chutes.

Length Wise - Assuming a 100% occupancy rate, about **14 Parcel** will readily fit into these chutes.

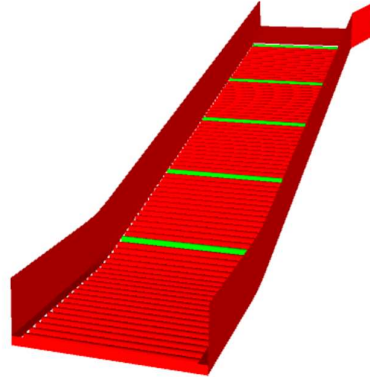
Each Chute is equipped with below components to ease the operations:

- Chute Full Sensors
- Three Color Tower Light
- Push Button

| Chute I/Os              | Description                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------|
| Chute Full Sensor       | To alert the system that the chute is full and mark that chute unavailable for further sortation |
| Three Colour Tower Lamp | To indicate the chute status                                                                     |
| Push Button             | To Start/Stop Sorting Operations                                                                 |

### 11.2.3.3 Rejection Chute (Friction Roller Type)

A Friction roller chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes its roller platform to gradually descend & collect the parcel at the end of the.



*Image for representation purpose only*

**Calculation for Design:** We have considered an average shipment size of 400 X 400 X 400 mm while designing the capacity of chutes.

| Average shipment size considered for chute volume calculation | L(m) | B(m) | H(m) | Volume (m3) |
|---------------------------------------------------------------|------|------|------|-------------|
|                                                               | 0.4  | 0.4  | 0.4  | 0.064       |
| Collection type chute                                         | 5.6  | 1    | 0.3  | 1.68        |

Volume Wise - Assuming a 100% occupancy rate, about **26 Parcel** will readily fit into these chutes.

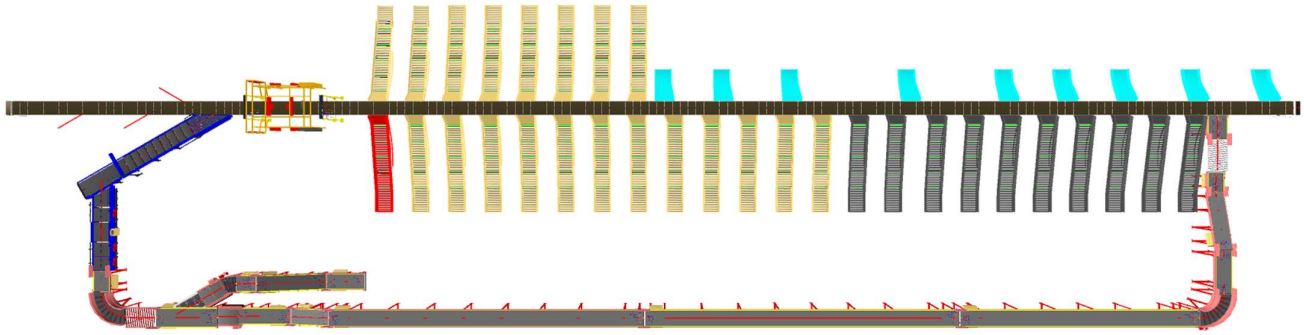
Length Wise - Assuming a 100% occupancy rate, about **14 Parcel** will readily fit into these chutes.

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensors
- Three Color Tower Light
- Push Button

| Chute I/Os                     | Description                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Chute Full Sensor</b>       | To alert the system that the chute is full and mark that chute unavailable for further sortation |
| <b>Three Colour Tower Lamp</b> | To indicate the chute status                                                                     |
| <b>Push Button</b>             | To Start/Stop Sorting Operations                                                                 |

#### 11.2.4 Recirculation & Manual Feed Line



**Recirculation Line:-** The recirculation line is strategically designed at the end of the Linear Cross-Belt Sorter (CBS) system to manage parcels that encounter sorting failures. This automated line efficiently gathers and transports the sort-failed parcels, refeeding them back into the Linear CBS system without requiring additional manual labor. The entire process is seamless, ensuring parcels are automatically re-fed into the sorting system for sorting.

**Manual Refeed Line:-** Additionally, this recirculation line is integrated to a manual feedline, enabling the reintroduction of the rejected parcels that have been manually reprocessed. This integration ensures that manually handled parcels are easily fed back into the Linear CBS, maintaining operational flow and minimizing delays.

##### 11.2.4.1 Conveyor BOQ

| S No. | Name                   | EL_1 | EL_2 | Length (m) | Width (mm) | MP (KW) | Set |
|-------|------------------------|------|------|------------|------------|---------|-----|
| 1     | PVC Belt Conveyor      | 1700 | 1700 | 3          | 1010       | 1.5     | 1   |
| 2     | PVC Belt Conveyor      | 1700 | 1700 | 4          | 1210       | 1.5     | 1   |
| 3     | PVC Belt Conveyor      | 2400 | 2400 | 15         | 1010       | 3       | 1   |
| 4     | PVC Belt Conveyor      | 2400 | 2400 | 20         | 1010       | 4       | 1   |
| 5     | PVC Belt Conveyor      | 2400 | 2400 | 20         | 1010       | 4       | 1   |
| 6     | PVC Belt Conveyor      | 1548 | 2482 | 4          | 1010       | 2.2     | 1   |
| 7     | PVC Belt Conveyor      | 698  | 1779 | 4          | 1010       | 2.2     | 1   |
| 8     | PVC Belt Conveyor      | 2302 | 1812 | 3          | 1010       | 2.2     | 1   |
| 9     | PVC Belt Conveyor (S3) | 1650 | 1650 | 1          | 1210       | NA      | 1   |
| 10    | PVC Belt Conveyor (S3) | 1650 | 1650 | 1          | 1010       | NA      | 1   |
| 11    | PVC Belt Conveyor (S3) | 800  | 800  | 2          | 1010       | NA      | 1   |
| 12    | PVC Belt Conveyor (S3) | 1700 | 1700 | 1          | 1010       | NA      | 1   |
| 13    | PVC Belt Conveyor (S3) | 2400 | 2400 | 3          | 1010       | NA      | 1   |
| 14    | PVC Belt Conveyor (S3) | 1700 | 1700 | 1          | 1010       | NA      | 1   |
| 15    | Buffer Conveyor        | 1700 | 1700 | 2          | 1010       | NA      | 1   |
| 16    | Buffer Conveyor        | 1700 | 1700 | 1          | 1010       | NA      | 1   |
| 17    | Belt Merge 30°         | 1700 | 1700 | 0          | 1010       | NA      | 1   |
| 18    | Belt Turn 90°          | 2400 | 2400 | 0          | 1010       | NA      | 1   |
| 19    | Belt Turn 90°          | 1700 | 1700 | 0          | 1010       | NA      | 1   |
| 20    | Belt Turn 30°          | 1700 | 1700 | 0          | 1010       | NA      | 1   |
| 21    | Aligning Conveyor      | 1700 | 1700 | 2          | 1200       | NA      | 1   |
| 22    | Aligning Conveyor      | 1650 | 1650 | 2          | 1200       | NA      | 1   |

## 12. Description of Components of Equipment

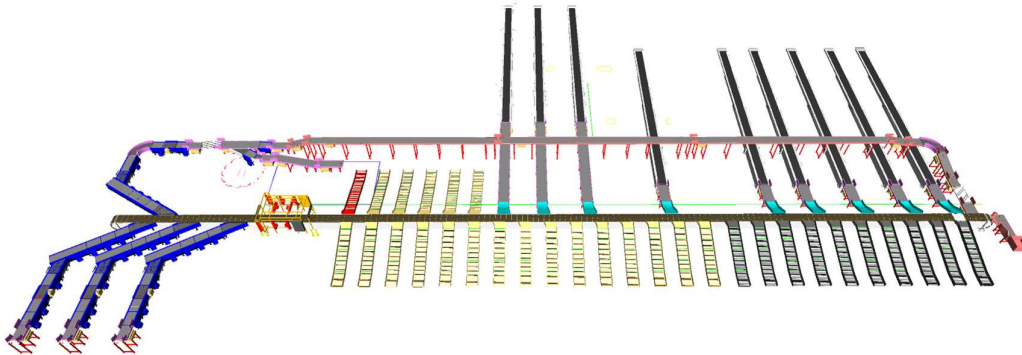
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

### 12.1 Cross Belt Sorter -

Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



#### 12.1.1 CBS Carrier

Falcon Autotech's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners" and non-sortable shipment recirculation.



### 12.1.2 Servo Roller

High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.



### 12.1.3 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.



### 12.1.4 Carrier Wheels

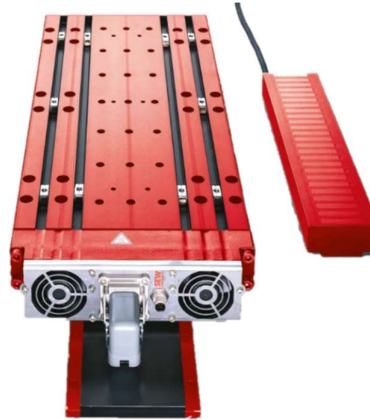
Carrier wheels that are thoroughly tested and proved for long life cycle.





#### 12.1.5 Non-contact based linear motor drive

Extremely High Energy Efficient Linear Synchronous Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed.



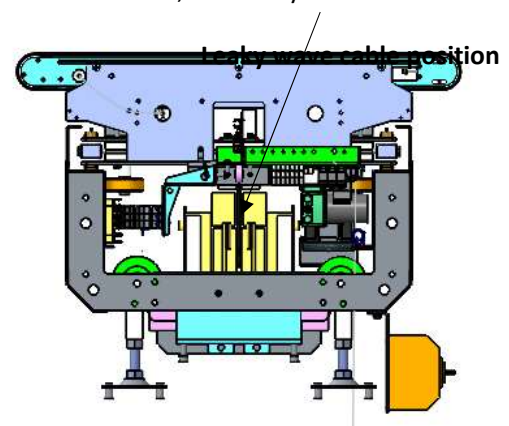
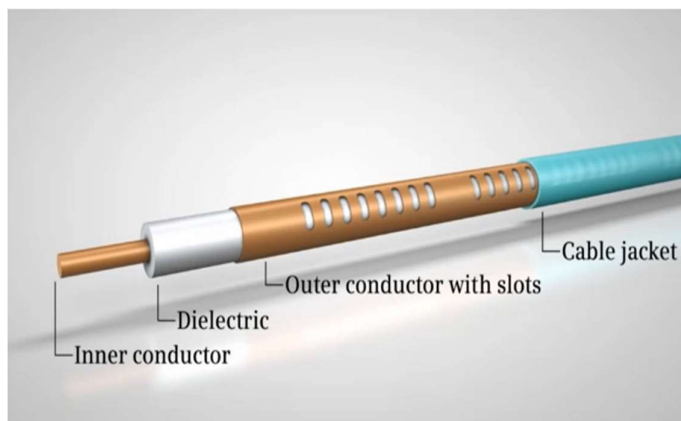
#### 12.1.6 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.



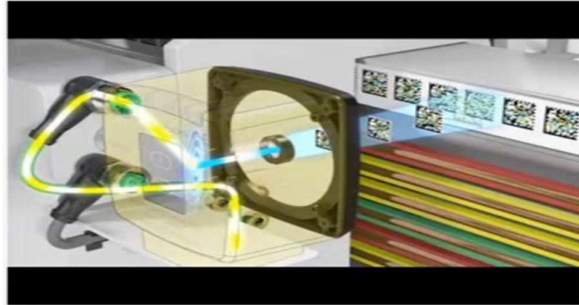
#### 12.1.7 Data Transmission

The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.



### 12.1.8 Carriers positioning System.

Positioning system determines the exact location of each carrier in the Linear at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter Linear, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



### 12.1.9 Technical specifications of Sorter

| Items                                     | Make                         |
|-------------------------------------------|------------------------------|
| Sorter Carrier Type                       | Linear CBS                   |
| Sorter Speed in m/s                       | Up to 1.5 m/s                |
| Sorter Linear Length (Top Side Only)      | ~81.58 m                     |
| Sorter Actuation Technology               | Electric                     |
| <b>Chutes</b>                             |                              |
| Live Chute                                | 09                           |
| Collection Chute                          | 20                           |
| Rejection Chute                           | 1                            |
| Recirculation Line                        | 1                            |
| RC Line                                   | 1                            |
| Sort fail Chute                           | 1                            |
| Height of Sorter                          | ~1700 mm                     |
| <b>Carrier Design</b>                     |                              |
| Carrier Motor Make                        | Falcon                       |
| Carrier Pitch - mm                        | 500                          |
| No. of Carriers                           | 333 Nos.                     |
| <b>Drives</b>                             |                              |
| Motor/Drive type                          | Linear Synchronous Motor     |
| Power Consumption                         | Will be specified during DAP |
| Feedlines (Fully Automatic)               | 06 Nos                       |
| <b>Power and Data Transmission</b>        |                              |
| Power Transmission                        | Over Continuous Bus Bar      |
| Maintenance Speed of the Sorter- Jog Mode | 0.8 m/s                      |
| Area Requirement (L x B)                  | Refer Layout                 |

## 12.2 Scanning & Sensing on Linear Sorter

### 12.2.1 Automatic Barcode Scanning

Top Sided Barcode Scanner to scan shipment 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time. As per AMAZON requirements, Scanning is in AMAZON scope.



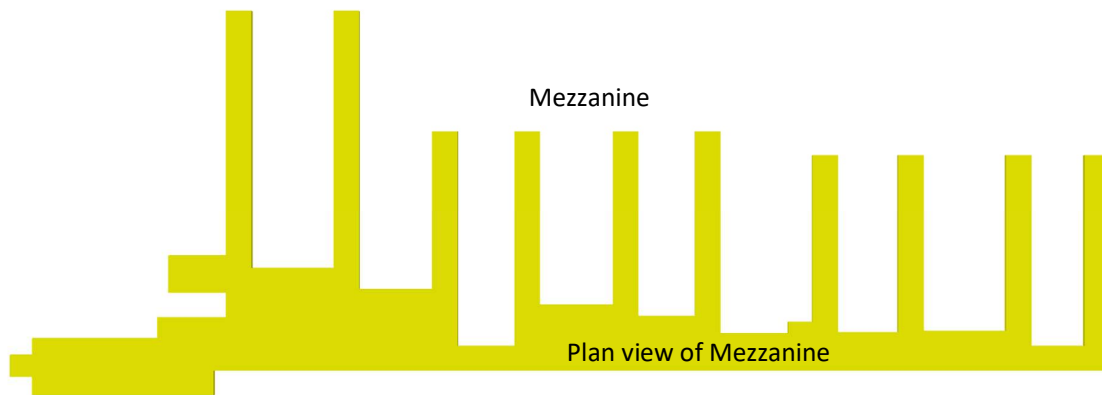
### 12.2.2 Shipment Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple shipment positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Shipment on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).



### 12.3 Steel Works – Mezzanine & Staircases- (In Amazon's Scope)

#### 12.3.1 Mezzanine



- Total Mezzanine Area: 2001 Sq. Metres
- Clear height below Mezzanine: 2100 mm
- No of Stairs: 12 Nos

#### 12.3.2 Staircase



- No of Stairs: 12 Nos

### 13. Proposed System Technical Details

#### 13.1 Mechanical equipment

| Pos. | Qty. | Description                                                                                                                                                                                                                                                                                                                | Value                                                                                       |
|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1    | 1    | <b>Infeed System</b> <ul style="list-style-type: none"> <li>Powered Belt Conveyors</li> <li>Aligning Conveyor</li> <li>Belt Turn 60 deg</li> <li>Belt Turn 90 deg</li> <li>Buffer Conveyor</li> <li>Angle Merge</li> </ul>                                                                                                 | ~774 metres (69 Modules)<br>~10 metres (05 Modules)<br>05 Nos<br>17 Nos<br>10 Nos<br>05 Nos |
| 2    | 6    | <b>Feedlines</b><br>Each fully Auto induct Consists of <ul style="list-style-type: none"> <li>Receiving Conveyor</li> <li>Buffer Conveyor</li> <li>Spacing Conveyor</li> <li>Weighing Conveyor</li> <li>Angle Merge</li> </ul>                                                                                             | 1 Set<br>10 Set<br>2 Set<br>1 Set<br>2 Set                                                  |
| 3    | 1    | <b>Sorter</b><br><b>1 Linear Cross Belt Sorter</b> <ul style="list-style-type: none"> <li>Sorter Height</li> <li>Sorter Length</li> <li>Sorter Speed</li> <li>Sorter Drive</li> <li>Carrier Pitch</li> </ul> <b>Including:</b> <ul style="list-style-type: none"> <li>Standard Sorter Supports</li> <li>Fencing</li> </ul> | 1700mm (from mezz level)<br>~81.5 m (Top Side)<br>~1.5 m/s<br>Linear Motor<br>500 mm        |
| 4    | 1    | Sorter Outputs <ul style="list-style-type: none"> <li>Live Chute</li> <li>Conveyors for Live Chutes</li> <li>Collection Chute</li> <li>Rejection Chute</li> <li>Sort fail Chute</li> </ul>                                                                                                                                 | 09 Nos<br>~50 m (10 Modules)<br>20 Nos<br>01 Nos<br>01 Nos                                  |
| 5    | 1    | <b>RC Line</b> <ul style="list-style-type: none"> <li>PVC Belt Conveyor</li> </ul>                                                                                                                                                                                                                                         | ~41 m (05 Modules)                                                                          |
| 6    | 1    | <b>Recirculation + Rejection Refeed Line</b> <ul style="list-style-type: none"> <li>PVC Belt Conveyor</li> <li>Aligning Conveyor</li> <li>Belt Merge</li> <li>Belt Turn 30 deg</li> <li>Belt Turn 90 deg</li> <li>Buffer Conveyor</li> </ul>                                                                               | ~81 m (14 Modules)<br>02 Nos<br>01 Nos<br>01 Nos<br>02 Nos<br>02 Nos                        |
| 7    | 1    | <b>Steel Works (In Amazon's Scope)</b> <ul style="list-style-type: none"> <li>Maintenance Platform</li> <li>Stairs</li> </ul>                                                                                                                                                                                              | ~879 SQM<br>12 Nos                                                                          |

### 13.2 Electrical Equipment

| Electricals |   |                                                                                                                                                                      |          |
|-------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1           | 1 | <b>Consists of</b><br><br>Main Power Distribution panel<br>Main Control Panel<br>Feedline Control Panels<br>Sorter Drive Panels<br>Network Switches<br>Field Cabling | Included |

### 13.3 Control System

| Components |   |                                                                                                    |                              |
|------------|---|----------------------------------------------------------------------------------------------------|------------------------------|
| 1          | 1 | <b>Consists of</b><br><br><b>Siemen's PLC based Control System with SCADA</b><br>Industrial Switch | 1 Nos.<br>As per requirement |

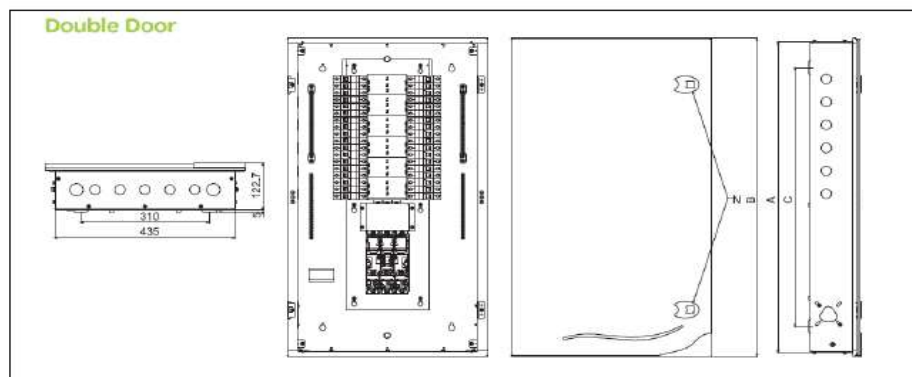
## 14. Electrical System

Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.

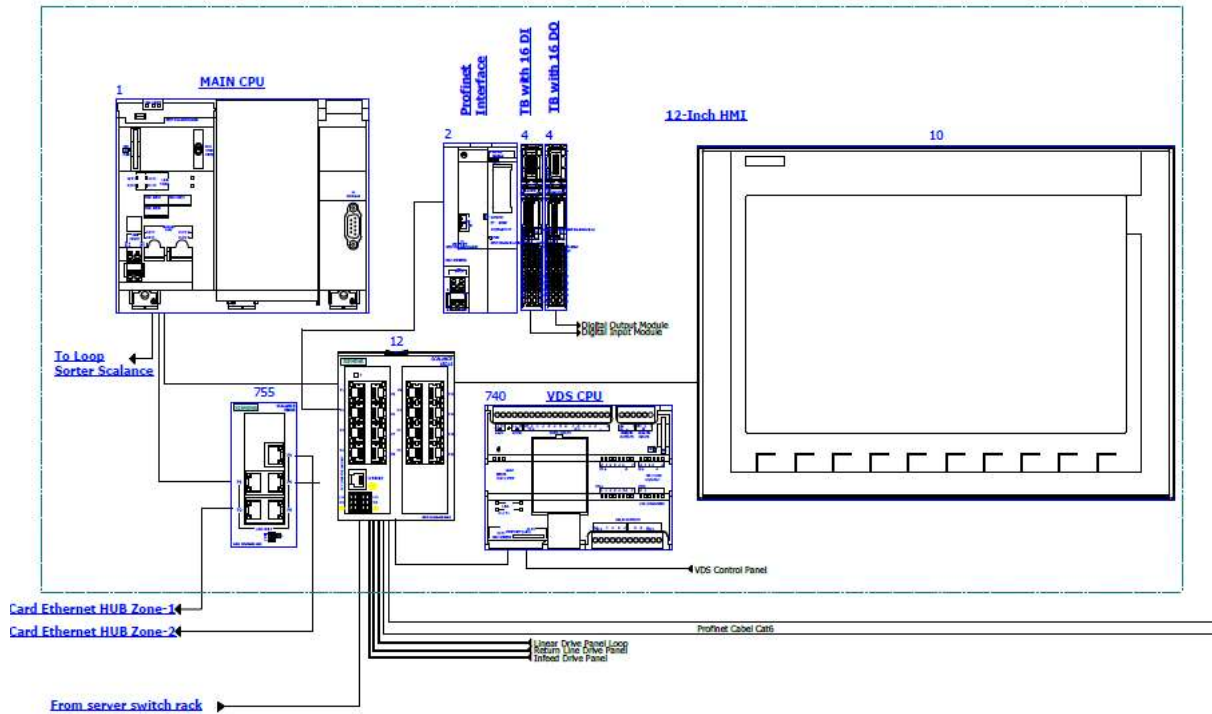
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

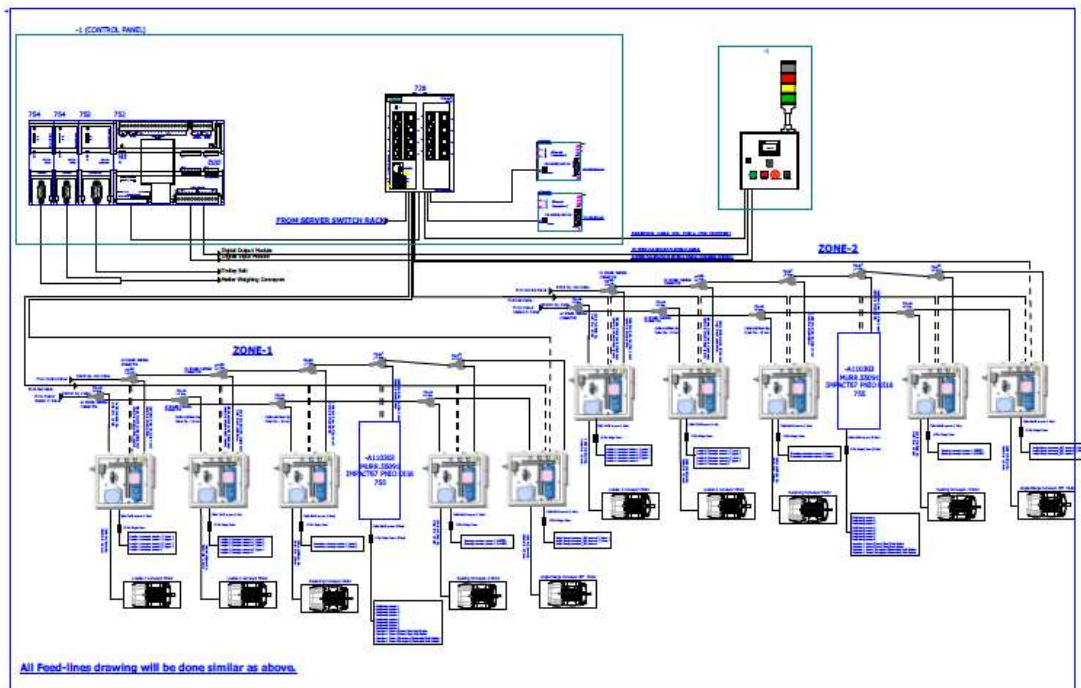
### 14.1 Reference Picture of Power Distribution Panel



## 14.2 Main Control Panel (Reference)



### 14.3 Induct Stations Control Panel (Reference)





## Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter.  
The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.  
All motors will have appropriate IP ratings

## Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

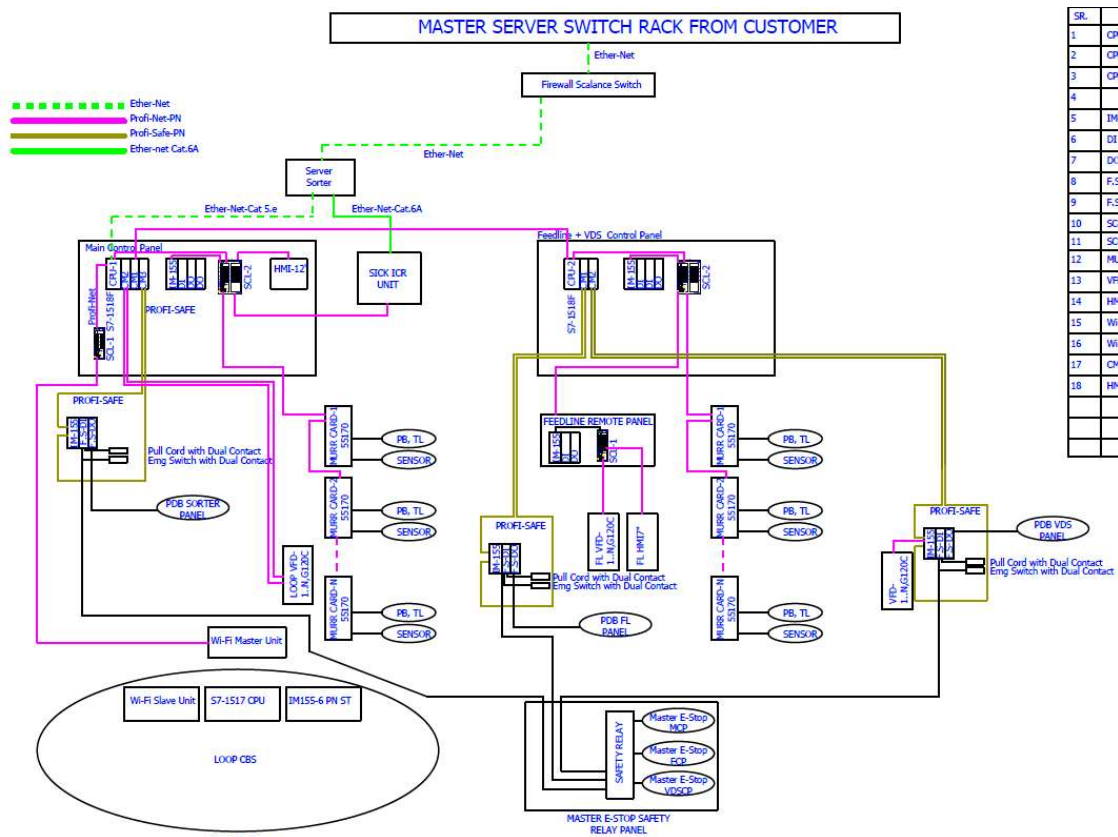
### Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Linear), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

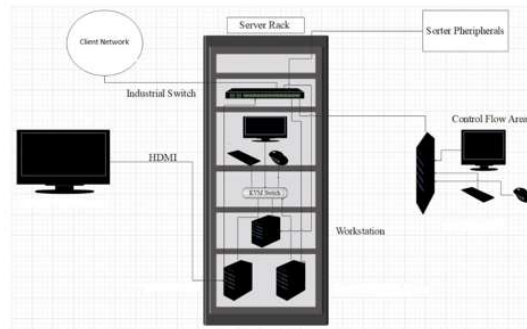
## Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

## 15. Controls Architecture (Ref.)



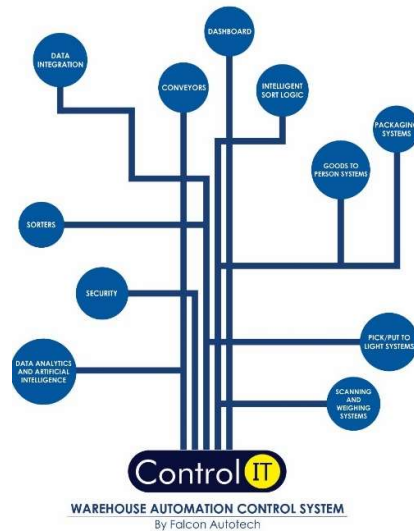
## 16. IT Architecture (Ref.)



### 16.1HAA Server (In Amazon Scope)

| S.no                                                              | Description                                                                                                                                                                                                                                                                                                                                                              | Qty |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>20Core Config with 128 GB RAM in T440 and 64 GB RAM in T40</b> |                                                                                                                                                                                                                                                                                                                                                                          |     |
| 1                                                                 | <b>Synology_storage_DS723+</b> 2.6 GHz AMD Ryzen R1600 Dual-Core, 2 x Gigabit Ethernet Ports,<br>2GB ECC DDR4 RAM, 2 x 3.5/2.5" Bays   2 x M.2 2280 Slots E10G22-T1-Mini 10GbE RJ-45 network upgrade module for compact Synology servers                                                                                                                                 | 2   |
| 2                                                                 | <b>Dell Tower Model T440</b> -PowerEdge T440 Xeon Gold 6148 2.4GHz/20C/27.5MB/150 W 16 DIMMS 4 x32GB RDIMM Up to 8 3.5"" Hot Plug Hard Drives Tower Configuration 2 x 1.2TB 10K RPM SAS 12Gbps 512n 2.5in<br><br>Hot-plug Hard Drive PERC H750 Adapter Full Height 8GB Cache Dual Hot-plug PS495W iDRAC9Enterprise 3YR ProSupport Next Business Day Onsite/Dual Port Lan | 2   |
| 3                                                                 | <b>Dell PowerEdge T40</b> Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T,Turbo,71W,TPM, 4*16 GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1Gbe LOM ,DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3<br><br>Year Onsite NBD,480 GB SSD,1 Gbe Dual Lan Card Extra                                                                               | 1   |
| 4                                                                 | Windows Server 2019                                                                                                                                                                                                                                                                                                                                                      | 3   |
| 5                                                                 | Monitor                                                                                                                                                                                                                                                                                                                                                                  | 1   |
| 6                                                                 | KVM Switch                                                                                                                                                                                                                                                                                                                                                               | 1   |
| 7                                                                 | Mouse                                                                                                                                                                                                                                                                                                                                                                    | 1   |
| 8                                                                 | Keyboard                                                                                                                                                                                                                                                                                                                                                                 | 1   |
| 9                                                                 | Lan Cable                                                                                                                                                                                                                                                                                                                                                                | 10  |
| 10                                                                | Power Cable                                                                                                                                                                                                                                                                                                                                                              | 8   |
| 11                                                                | VGA Cable                                                                                                                                                                                                                                                                                                                                                                | 1   |
| 12                                                                | 42 AC Server Rack                                                                                                                                                                                                                                                                                                                                                        | 1   |
| 13                                                                | Netgear 24 Port Giga Switch                                                                                                                                                                                                                                                                                                                                              | 2   |
| 14                                                                | 2 TB SSD Micron                                                                                                                                                                                                                                                                                                                                                          | 4   |
| 15                                                                | DP to VGA Convertor Cadyce Brand                                                                                                                                                                                                                                                                                                                                         | 1   |

## 17.Falcon's WCS CONTROLIT



### Key Values

#### Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

#### Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

#### Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

#### Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

#### Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

### Supported Protocols

|             |                  |                         |                       |
|-------------|------------------|-------------------------|-----------------------|
| API         | FTP              | FILE UPLOAD / DASHBOARD | ODBC                  |
| WEBSERVICES | AMAZON S3 SERVER | FIREBASE                | Any Customer Protocol |

## 18.Falcon's Visual Inspection System (SCADA)

SCADA stands for Supervisory Control and Data Acquisition. It is a system of hardware and software components that allows for remote monitoring, control, and data acquisition of industrial processes or facilities

The Visualization system provided by FALCON (or SCADA) allows the monitoring and control of the different systems delivered for the Amazon Riyadh Hub. This SCADA system receives from each monitored sub-system all information on their operating status in real time.

At the system monitoring level, the functions performed are:

- Field data acquisition.
- Animated visualization of equipment.
- Representation of the operating mode of the system (nominal, contingency, etc.).
- Alarm management.
- Alarm history management.
- Diagnostic help.
- Failure detection.
- Equipment control.
- Statistics on equipment operation.
- Historical Statistical Report.
- Recording and archiving.
- Safety operator interface.

## 19.Key Components Make

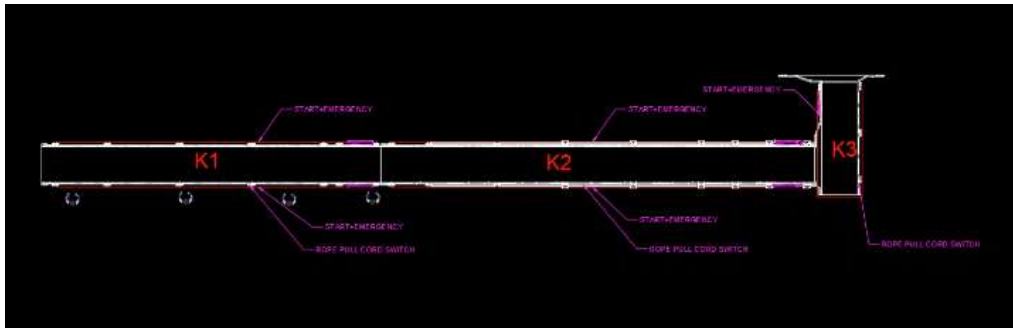
| Items                             | Make                          |
|-----------------------------------|-------------------------------|
| <b>Belts</b>                      | Forbo/ Derko                  |
| <b>Rollers</b>                    | Falcon                        |
| <b>Cross Belt Carriers</b>        | Falcon                        |
| <b>Linear Synchronous Motors</b>  | Falcon/SEW                    |
| <b>Feed Line Motors</b>           | Induction Motors – SEW/ Lenze |
| <b>Weighing Conveyor</b>          | Metter/Beizerba               |
| <b>Barcode Scanner</b>            | SICK                          |
| <b>Encoders</b>                   | SICK                          |
| <b>Sensors</b>                    | Sick/Leuze/ P&F               |
| <b>PLC</b>                        | Siemens                       |
| <b>Control Panels</b>             | Rittal/ BCH                   |
| <b>VFDs</b>                       | Siemens/Lenze/ AB/ Omron      |
| <b>Cables</b>                     | LAPP/ Equivalent              |
| <b>Switch Gear</b>                | Schenider/Equivalent          |
| <b>Bearings</b>                   | NTN/SKF/Equivalent            |
| <b>Power Transmission Systems</b> | Vahle                         |
| <b>HMIs</b>                       | Siemens                       |

## 20.Principle of Safety

### 20.1E-Stops



- a. At Every Conveyor Module – Both sides



## 21. Program Organisation

### 21.1 Program/ Project Schedule

| S.No. | Activity                    | Timeline |
|-------|-----------------------------|----------|
| 1.    | Receipt of PO               | Week 0   |
| 2.    | Project Kick off            | Week 1   |
| 3.    | DAP Submission              | Week 4   |
| 4.    | DAP Approval                | Week 6   |
| 5.    | Detailed Designing          | Week 10  |
| 6.    | Manufacturing & Procurement | Week 18  |
| 7.    | Inhouse Assembly            | Week 20  |
| 8.    | Dispatch                    | Week 21  |
| 9.    | Installation & Go Live      | Week 26  |

*\*Above plan is tentative, detailed plan will be shared post order confirmation.*

### 21.2 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Amazon
3. Weekly/Fortnightly Meeting Amazon to share the Project Status.
4. Scheduling of the resource management.
5. Management of risks and opportunities.
6. Management of the requirements.
7. Management of the list of anomalies or reservations.

## 22. Amazon Responsibility

### 22.1 Amazon Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- Amazon to provide HPT/ Forklift for material movement for complete duration.
- The possibility of authorizing access to the site and the execution of the installation work up to 7 days a week and 24 hours a day if deemed necessary and if requested

by FALCON.

- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the operation of the shipmentsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.
- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

## 22.2Responsibilities of Amazon During the Tests

- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the shipments correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

## 22.3Amazon Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough spacefor desks or tables and chairs for the trainer and trained staff.
- check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technicalstaff.
- The invitation of staff to attend the training courses will be at the expense of AMAZON.



## 23.Commercial Terms

### 23.1Price Sheet

| S. No             | Component                    | Set | Price (INR) |              |
|-------------------|------------------------------|-----|-------------|--------------|
| 1                 | Linear Cross Belt Sorter     | 1   | ₹           | 8,37,91,942  |
| 2                 | Conveyor Automation          | 1   | ₹           | 8,50,76,219  |
| 3                 | Output Chutes                | 1   | ₹           | 78,13,300    |
| 4                 | Software Packages            | 1   | ₹           | 11,00,000    |
| 5                 | Installation & Commissioning | 1   | ₹           | 88,34,073    |
| 6                 | Packaging & forwarding       | 1   | ₹           | 35,33,629    |
| 7                 | Project Management           | 1   | ₹           | 47,53,729    |
| 8                 | Freight                      | 1   | ₹           | 44,80,000    |
| Grand Total (INR) |                              |     | ₹           | 19,93,82,892 |

The Total Price is to be understood.

- Freight - Included
- Taxes as applicable
- Packing and Forwarding - Included
- Price is valid for 60 days from the date of proposal.

\*Material Unloading & Laydown area is in customer scope

### 23.2Payment Terms

50% Advance with PO.

30% With readiness of material before dispatch.

20% After successful installation & Commissioning

*\*All invoices (if due as per payment terms) should be cleared within 7 days from the date of receipt of undisputed invoice by the buyer*

## 24. Warranty Period

Falcon's offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use).

The warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in the Exclusion Clause)

The warranty does not apply to the replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
  - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
  - Negligence or abnormal use of equipment.
  - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
  - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

## 25. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Server System**
- **Mobile Carts**
- **Construction Power**
- **Steel Works**
- **Pallets at chutes**
- **PC for SCADA.**
- **Stress free mats**
- **Insulation mats**
- **Fans at chutes & inducts.**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.